Materials for Control Valves

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Authors:

Don Bush
Jim Gossett
Ted Grabau

Materials Engineering Group
Fisher Controls International, Inc.
Marshalltown, Iowa

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MATERIAL PROPERTIES **4**
MECHANICAL AND PHYSICAL PROPERTIES 4
WEAR PROPERTIES 5
ENVIRONMENTAL CONSIDERATIONS **6**
COMMON MATERIAL/ENVIRONMENT COMPATIBILITY CONSIDERATIONS IN CONTROL VALVES 6
Erosion Corrosion *7*
Environmentally-Assisted Failure *7*
Stress Corrosion Cracking (SCC) 7
Hydrogen Damage 7
Hydrogen Embrittlement 7
Hydrogen Attack 8
Hydrogen Blistering 8
Liquid-Metal Embrittlement (LME) 8
Solid Metal Induced Embrittlement (SMIE) 8
Crevice Corrosion *8*
Pitting *8*
Intergranular Corrosion *8*
Galvanic Corrosion *9*
Selective Leaching *9*
Chemical Compatibility of Non-Metallics *9*
TEMPERATURE EFFECTS 9
Effects of Elevated Temperature on Metallurgical Stability *9*
Effects of Elevated Temperature on Yield Strength *10*
Creep at Elevated Temperature *10*
Effects of Elevated Temperature on Elastic Modulus *10*
Coefficient of Thermal Expansion *10*
Effects of Low Temperature on Toughness *11*
SPECIFIC MATERIAL/ENVIRONMENT CONSIDERATIONS AND LIMITATIONS 11
Gray cast iron and many of its variations: *11*
Carbon and Alloy Steels: *11*
Stainless Steels: *12*
Nickel Alloys: *12*
Copper Alloys: *12*
Miscellaneous: *12*
MATERIALS OF CONSTRUCTION **13**
VALVE BODIES AND BONNETS 13
Codes for Pressure Boundary Parts *13*
ANSI/ASME B16.34, *Valves - Flanged, Threaded, and Welding End*: 13
ASME Boiler and Pressure Vessel Code, Section VIII: 13
ANSI/ASME B31.1, *Power Piping Code*, ANSI/ASME B31.3, *Chemical Plant and Petroleum Refinery Piping Code*, and ANSI/ASME B31.5, *Refrigeration Piping Code*: 13
ANSI/ASME B16.1, *Cast Iron Pipe Flanges and Flanged Fittings* and ASME B16.42, *Ductile Iron Pipe Flanges and Flanged Fittings*: 14
Standard Material Specifications for Pressure Boundary Parts *14*
Materials for Pressure Retaining Parts *15*
Bodies and Bonnets 15
Gray Cast Iron 15
Ductile Cast Iron 15
Carbon Steels 15
Alloy Steels 16
Ferritic Stainless Steels 16
Martensitic Stainless Steels 16
Austenitic Stainless Steels 16
Super-Austenitic Stainless Steels 17
Duplex Stainless Steels 18
Nickel Alloys 18
Titanium, Zirconium, and Tantalum 18
Copper Alloys 19

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Bolting 19
Grades B7 and L7 Bolts and Grades 2H and 7 Nuts 19
Grade B16 19
Grade B8M 19
Coatings 20
Electroless Nickel Coating (ENC) 20
Aluminizing 20
Boronizing 20
Sprayed Coatings 20
Polymeric Liners 20
TRIM MATERIAL SELECTION 21
Environmental Considerations *21*
Mechanical and Physical Properties *21*
Materials of Construction *21*
Globe Valve Components: 22
Plugs 22
Seat Rings 22
Cages 22
Bushings 22
Stems 22
Materials-Related Problem Areas in Globe Valve Trim: 23
Plug O.D. and Seat Line Erosion 23
Plug/Seat Interface Erosion (Wire Drawing) 23
Cage Opening Erosion (Wire Drawing) 23
Cavitation Damage on Plugs, Seat Rings, and Cages 23
Plug/Cage Interface Galling 24
Port-Guided Plug/Seat Interface Galling 24
Post-Guided Plug/Bushing Interface Galling 24
Plug/Stem Connection Failure 24
Common Globe Valve Trim Material Combinations: 25
Rotary Valves Components: 26
Disk/Ball/Plug 26
Seal/Seat 26
Shaft 26
Bearings 26
Pins 26
Materials-Related Problem Areas in Rotary Valve Trim: 27
Ball/Shaft Connection Failure 27
Ball and Seal Wear 27
Shaft/Bearing Wear and Galling 27
Common Rotary Valve Trim Material Combinations: 27
VALVE PACKING 28
PTFE V-Ring Packing: *28*
Graphite/Carbon Packing: *29*
GASKETS 30
Elastomeric Gaskets *30*
PTFE Gaskets *30*
Asbetos Gaskets *30*
Aramid Gaskets *30*
Metal Gaskets *31*
Flexible Graphite Gaskets *31*
O-ring Seals *31*
Spring Energized, Pressure Assisted Seals *31*
SEALANTS 32
Polymeric Adhesives *32*
Metallic Dispersions *32*
PAINT AND EXTERNAL COATINGS 33
Pretreatment *33*
Alkyds *33*
Acrylic Latex *33*
Epoxies and Polyesters *34*
TRADEMARKS **34**

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The selection of materials for control valve components is a very complex undertaking. Control valves are required to function with precision in some very extreme environments. A number of factors must be considered to insure that a material will perform properly in service. These factors fall primarily into two categories:

1. the material's suitability to function mechanically, and
2. the material's compatibility with the environment.

To make matters difficult, these categories conflict in many instances, making it difficult or impossible to satisfy all considerations with a single material. In these cases, the best compromise must be identified.

Material Properties

Mechanical and Physical Properties

When selecting materials, the mechanical and physical properties which must be considered vary depending upon the component. Obviously, the properties which are important in the selection of a body material are different from those used in the selection of trim material. Some of the properties which must be considered when selecting valve materials are described below:

Elastic Modulus: In metallic materials, stress (S = load divided by area) is proportional to strain (e = change in length divided by initial length) provided the stress is below a threshold stress, called the yield stress, where permanent (plastic) deformation begins to occur. The elastic modulus (E) relates stress and strain by the equation:

$$S = E \cdot e$$

The elastic modulus is basically a measure of the "stiffness" or "spring rate" of the material, and is only dependent upon composition and temperature.

Tensile Strength: The tensile strength is the stress required to cause rupture. Tensile strength is not generally used directly in design, since it is seldom desirable to utilize a component in a situation where it is on the verge of failure. However, the tensile strength value is utilized in the computation of allowable stresses in most codes.

Yield Strength: The yield strength of a material is the stress required to cause a permanent deformation of 0.2%. This parameter is also utilized in the computation of allowable stresses in most codes. It is generally a critical factor considered when selecting materials for parts which carry loads, such as valve stems, cages, seat rings, bolting, etc.

Hardness: Hardness is defined as a material's resistance to penetration, indentation, or scratching, and is one of the most difficult material properties to fully understand. In metals it is usually measured by loading an indenter into the material and measuring either the depth of penetration or the surface area of the indentation. The deeper the penetration or the greater the surface area of the indentation, the lower the hardness. Thus, the hardness as measured in this manner is a function of a number of other properties, such as yield strength, work hardening rate, elastic modulus, etc.

There is a general impression that hardness is directly related to the service life of a trim component, and that the hardness levels of two materials can be used to compare their "value" (hardness / dollar). However, the use of hardness as a gauge of wear resistance, erosion resistance, cavitation resistance, or galling resistance is merely a first-order approximation. There are a number of other material characteristics which contribute to resistance to these types of wear. The composition and crystal structure of a material, which are strongly related, can have a much greater effect than the actual hardness. This is the reason that cobalt-base hardsurfacing materials are superior in most wear situations, even though their hardness is relatively the same as for hardened stainless steels. It has been shown that the reason for the excellent performance of cobalt-base alloy 6 in wear applications is the crystal structure of its soft matrix phase, not its average

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hardness or its very hard carbide phase. A good rule of thumb to follow is that hardness can be used to compare alloys which are in the same alloy family (such as one 400 series stainless steel vs. another), but it is not very meaningful when used to compare alloys or materials with much different chemistries.

Toughness: Toughness is a material's resistance to fracture. Toughness has traditionally been measured using impact tests, such as the Charpy and Izod tests. These both measure the amount of energy (usually foot-pounds or joules) required to fracture a specimen with a pre-existing stress riser. Recently, the science of fracture mechanics has introduced new methods for both determining a material's resistance to fracture and evaluating a structure's susceptibility to fracture in the presence of defects. The measure of toughness in the fracture mechanics realm is called fracture toughness, and is a measure of the stress at the tip of a sharp crack that is sufficient to cause catastrophic failure in a particular material.

Fracture toughness values, and even Charpy and Izod impact toughness values, are difficult to find for many of the materials used in control valves. In most cases, they are even more difficult to correlate with operating conditions in a control valve. For these reasons, the other mechanical properties are often examined instead to give an indication of toughness. In general, a tough material displays a higher percent elongation and/or percent reduction in area than a brittle material. Also, a tougher material will display a greater difference in yield strength and ultimate tensile strength (or, a greater work hardening rate) than a brittle material. And finally, austenitic materials (such as 300 series stainless steels and nickel-base alloys) generally have much greater toughness than ferritic materials (such as carbon and alloy steels and 400 series stainless steels).

Wear Properties

Wear is a term used in conjunction with a number of mechanisms involving material removal or damage. The most important specific wear categories encountered in control valves are sliding wear, erosion, and cavitation damage.

Sliding wear refers to the damage caused when two mating parts move relative to one another. Sliding wear actually encompasses a number of different mechanisms. The two mechanisms most often encountered in metallic components of control valves are adhesive wear and oxidative wear.

Adhesive wear (usually called "galling"), occurs when the frictional heat and contact pressure between asperities (small irregularities) on the surfaces of two parts are sufficiently high to cause localized welding. The relative motion of the parts causes repeated welding and fracture of these localized areas, causing material transfer between the parts. The surfaces of both parts become rough, which in most cases aggravates the situation. The roughness of the parts reduces mechanical efficiency, and can even cause complete seizing of the parts.

Oxidative wear is similar to adhesive wear, except that the frictional heat causes oxidation of the asperities. Oxidative wear generally produces a fine, powdery wear product, which may or may not cause abrasive damage to the metallic parts. Whether sliding wear is adhesive or oxidative in nature depends on a number of factors, including the wear couple materials, the contact pressure, and the environment. Galling is more likely to occur in inert atmospheres, whereas oxidative wear is more likely in atmospheres which are reactive toward the metal alloys involved.

It is often stated that sliding wear resistance can be optimized by following several guidelines:

- Use of mating materials with dissimilar elemental composition, which makes welding of the materials at the wear interface less likely. This is sometimes accomplished through the use of plating, diffusion coatings, or weld overlays.
- Use of materials with different surface hardness. This is also sometimes accomplished through the use of plating, diffusion coatings, or weld overlays.

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- Use of lubricants where possible. Lubricants reduce frictional heating and interfere with welding of the materials at the interface.

In most cases, factors such as corrosion, erosion, or strength considerations will limit the number of candidate materials and prevent these wear guidelines from being followed. In these instances, the component materials providing the best combination of properties must be determined.

Erosion is mechanical damage caused by either high-velocity fluid impingement or impact by abrasive particles in the flow medium. Erosion-corrosion is the combined effect of erosion and corrosion, and is commonly encountered in control valves. Cavitation damage is caused by the shock waves generated when vapor bubbles implode during pressure recovery. Erosion, erosion-corrosion, and cavitation damage can be minimized by material selection, although in most cases the use of an appropriate valve and/or trim style is more effective.

Environmental Considerations

A number of environmental factors influence the selection of control valve materials. In general, material/environment compatibility (such as corrosion, embrittlement, etc.) and temperature effects (such as stress relaxation and creep) are the predominant considerations.

Common Material/Environment Compatibility Considerations in Control Valves

There are a number of general material/environment compatibility considerations which should be evaluated when selecting materials and construction techniques for control valves:

General Corrosion

Although the vast majority of control valves are sold with carbon steel valve bodies, indicating they are intended for applications which require minimal corrosion resistance, corrosion is often a consideration. The topic of corrosion is obviously too broad to be fully covered here. However, some general guidelines will be provided.

Table 1 provides information concerning the general corrosion resistance of common valve materials in a number of environments. These types of tables are commonly included in manufacturers' literature to help customers select materials of construction. Unfortunately, due to space constraints, these tables only provide a general indication of how various materials will react when in contact with certain fluids at ambient temperature. The data cannot be absolute because concentration, temperature, presence of impurities, pressure and other conditions may alter the suitability of a particular material. Another limitation of these tables is that they do not usually provide information for corrosion types other than general corrosion, which involves uniform material removal over the entire exposed surface. There are also economic considerations that may influence material selection. Therefore, one should use this table as a guide only.

In general, when a valve is destined for a corrosive application, it is best to utilize quantitative corrosion data as a basis for material selection. Quantitative data, in the form of tables or iso-corrosion diagrams, is often available from the major material producers for corrosion-resistant alloys or in published corrosion data compilations. The tables and/or diagrams provide corrosion rate data, usually in mils per year (abbreviated "mpy"), as a function of concentration and temperature. One mil per year equates to the loss of 0.001" from the surface of an exposed part during one year of exposure. Some sources are not completely quantitative, but provide performance categories such as "less than 2 mpy", "2-20 mpy", "20-50 mpy", and "over 50 mpy". If possible, it is generally best to produce trim from materials that experience very low corrosion rates, since material loss will usually result in poor valve performance. Bodies are sometimes produced from materials that suffer slightly more corrosion, since they may continue to serve their purpose even after measurable corrosion damage has occurred.

There are seven other forms of corrosion which can cause problems in control valve applications:

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Erosion Corrosion

Erosion corrosion is a form of material removal involving the combined, synergistic effects of flow erosion and corrosion. Removal rates are dependent upon many factors, including the corrosive nature of the fluid, velocity, angle of impingement, size and shape distribution of entrained solid particles, and the mechanical properties of the metal. The most common form of erosion or erosion-corrosion in valves is "wire drawing", which is localized erosion or erosion-corrosion damage to seating surfaces or cage holes. Since trim parts are often exposed to higher local velocities than valve bodies, they must often be produced from materials that are more corrosion resistant than the body to avoid erosion-corrosion problems.

Cavitation damage is usually categorized as a form of erosion corrosion, although it involves the additional mechanical action of imploding vapor bubbles to produce material damage. The material removal rates due to cavitation are heavily dependent upon the corrosive nature of the fluid and the mechanical properties of the base material.

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Environmentally-Assisted Failure

Environmentally-assisted failure is a general term used to describe a number of processes which cause catastrophic failure of susceptible materials in particular environments. Environmentally-assisted failure encompasses a number of specific failure modes, including stress corrosion cracking. hydrogen damage, liquid-metal embrittlement, and solid metal-induced embrittlement.

Stress Corrosion Cracking (SCC)

SCC is environmentally-assisted, catastrophic cracking of a susceptible material in a particular environment. This phenomenon can produce cracking at stress levels well below a material's rated tensile strength. Stress corrosion cracking failures usually, but not always, display multiple, branched cracks. The major factors that influence SCC include material condition, environmental composition and temperature, and stress level. Specific examples of material / environment combinations that can cause SCC are covered in the "Specific Material / Environment Considerations and Limitations" section of this chapter.

Hydrogen Damage

Hydrogen damage is a term that encompasses a number of hydrogen-related failure modes, including hydrogen embrittlement, hydrogen attack, hydrogen blistering, and several other phenomena that are uncommon in the valve industry.

Hydrogen Embrittlement

Hydrogen embrittlement, also called hydrogen stress cracking or hydrogen-induced cracking, is a condition of low ductility in metals resulting from the absorption of hydrogen. Hydrogen embrittlement is mainly a problem in steels with ultimate tensile strength greater than 90 ksi, although a number of additional alloys are susceptible. Most hydrogen embrittlement failures occur as a result of absorption of hydrogen that is generated during plating, pickling, or cleaning operations. However, hydrogen charging may also occur in-service, especially in cases where hydrogen is generated due to corrosion. Hydrogen embrittlement failures are generally characterized as delayed, catastrophic failures occurring at stresses below the yield strength, and exhibiting single, non-branching cracks. Cracking of materials in hydrogen sulfide environments, usually called sulfide stress cracking (SSC) or wet H_2S cracking, is a special case of hydrogen embrittlement wherein the H_2S dissociates into hydrogen and sulfide ions in the presence of water, and the sulfide ions catalyze the absorption of hydrogen into the susceptible material. Selection of materials for H_2S environments is generally based upon NACE standard MR0175, *Sulfide Stress Cracking Resistant Metallic Materials for Oilfield Equipment*. Materials listed in MR0175 have demonstrated satisfactory performance in field exposure and / or laboratory tests. Many valves sold for oil production applications, oil and gas refineries, and other applications where H_2S and water are present, are built per NACE MR0175 requirements.

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Hydrogen Attack

When carbon and low-alloy steels are exposed to high-pressure, high-temperature hydrogen, the hydrogen will diffuse into the steel and combine with the carbon in the steel to form methane gas, which causes decarburization and / or internal cracking, resulting in reduced strength. Resistance to hydrogen attack increases with increasing chromium and molybdenum levels, since these elements form more stable carbides than iron, and do not release the carbon to the hydrogen as readily.

Hydrogen Blistering

Hydrogen blistering is the formation of blisters containing hydrogen gas in steels. This occurs when atomic hydrogen diffuses through the steel and recombines into molecular hydrogen (H2) at internal defects, such as voids, laminations, and non-metallic inclusions. The newly-formed molecular hydrogen cannot diffuse back out through steel, so build-up of molecular hydrogen results in increased pressure inside the defect cavities, eventually causing blistering of the material. Killed steels are often specified for hydrogen-containing processes because they are more resistant to hydrogen blistering than rimmed or semi-rimmed steels due to the relative lack of internal voids.

Liquid-Metal Embrittlement (LME)

Liquid-metal embrittlement occurs when a normally ductile metal cracks in tension when in direct contact with another metal that is in the liquid form. As is the case with stress corrosion cracking, there are particular metal / liquid metal couples that are known to exhibit liquid metal embrittlement. The main factors that affect LME include temperature, material strength level, and applied / residual stress level. LME is characterized by catastrophic failure at stresses below the yield strength. LME fracture surfaces generally consist of a single crack with complete coverage by the liquid metal.

Solid Metal Induced Embrittlement (SMIE)

Solid metal induced embrittlement occurs when metal / metal couples display embrittlement below the melting point of the lower-melting material. The couples which experience SMIE also display liquid metal embrittlement, but not all LME couples will experience SMIE. Factors affecting SMIE include temperature, material strength level, and applied / residual stress level. SMIE fractures generally consist of multiple, intergranular initial cracks, with a final fracture that is ductile in nature.

Crevice Corrosion

Corrosive ions can concentrate in crevices, such as in tight clearances between the valve body and trim parts, in socket weld joints, and other stagnant, confined areas, causing increased corrosion in those areas. Crevice corrosionis a major problem in chloride environments, because chloride ions in crevices break down the metal's protective oxide layer.

Pitting

Pitting is a self-initiating form of crevice corrosion. This is an extremely localized attack that causes small holes in the metal. Pitting will initiate at thin or weak areas in the protective, oxide layer on stainless steels.

Intergranular Corrosion

Intergranular corrosion is corrosion occurring primarily in the grain boundaries. Material loss can be very rapid because grains are undermined, causing them to drop out. Intergranular corrosion is most commonly seen in the weld heat-affected zone of stainless steelsThe corrosion results from chromium carbide precipitation or "sensitization" in grain boundaries. Loss of chromium in the matrix lowers corrosion resistance in areas immediately adjacent to the grain boundaries. Corrosion proceeds around the grains, causing them to "drop out". Overall material loss can occur at very high rates.

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Galvanic Corrosion

When two dissimilar metals contact each other in the presence of an electrolyte (such as a process fluid), a battery effect is created, and current flows from one metal to the other. One metal becomes the anode, the other the cathode, and corrosion occurs on the anodic metal. An example is a steel valve body installed in a stainless steel piping system. If the environment is aggressive enough, the relatively small steel body will corrode at a rate much higher than it would if it were not in contact with the stainless steel piping.

Selective Leaching

This corrosion mode involves selective removal of one element from an alloy by chemical action. Common examples are leaching zinc from brass (dezincification) and iron from cast iron (graphitization).

Chemical Compatibility of Non-Metallics

The chemical compatibility of elastomers, plastics, and other non-metallic materials is fairly complex. These materials can undergo a number of changes when exposed to particular environments, causing phenomena such as swelling, shrinkage, dissolution, chain scission, hardness changes, loss of mechanical properties, etc. Prediction of these responses is practically impossible and actual testing must be done to qualify compatibility. Most compatibility testing is done via soak testing per ASTM D471 or similar method. Change in mass, volume, hardness and / or tensile strength and elongation are the usual indices of evaluation. Obviously, minimal changes after exposure connote compatibility with the environment.

Three tables are listed for elastomer (rubber) performance. Table 2 lists the usual industrial process engineering elastomers. This table is useful for screening candidates for a given application.

Table 3 is a more complete fluid compatibility table. This table rates and compares the compatibility of elastomers with specific fluids. The tabulation is based on published literature of various polymer suppliers and rubber manufacturers, laboratory tests, and records of actual service performance. Note that this information should be used as a *general guide only*. An elastomer which is compatible with a fluid may not be suitable over the entire range of its temperature capability. In general, chemical compatibility decreases with an increase in service temperature. Selection of an elastomer for a butterfly valve seat or liner can only be as accurate as the information on which it is based. Known factors should include: (1) temperature, (2) pressure, (3) all chemicals, primary as well as trace, in the flowing fluid, (4) rate of flow, and (5) type of valve action, throttling or on / off.

The chart in figure 1 shows the usable temperature ranges of fabric reinforced diaphragms. Note that this full temperature range does not apply to all environments. These temperature ranges are generally for air and other environments that are compatible with the elastomer and fabric. The temperature ranges may be limited by the elastomer or the fabric even though logical composite combinations have been listed. Other combinations are possible and must be evaluated based upon application, cost and performance.

Temperature Effects

Temperature excursions from ambient result in several changes in material properties which can affect performance of control valves. Some of the more prevalent effects are summarized below.

Effects of Elevated Temperature on Metallurgical Stability

Most metal alloys have structures that are metastable in nature, and when they are placed into an elevated temperature environment, they tend to transform to their stable structures. The reactions that occur can affect a number of properties. For example, carbon steel materials that are used for valve bodies possess a two-phase microstructure consisting of ferrite (essentially pure iron) and iron carbides. Prolonged exposure above 800°F (427°C)causes the carbides to decompose into iron and graphite, reducing both the strength and toughness of the material, a phenomenon known as "graphitization". Steel alloys with chromium and / or molybdenum are utilized above 800°F (427°C) because of their more stable carbide phases.

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Metallurgical stability problems at high temperatures affect other materials, and account for upper service temperature limits in many cases. Some examples of materials limited by elevated temperature stability problems include:

- S17400 and related precipitation hardenable stainless steels lose toughness when used at temperatures above 600°F (316°C). Since the toughness reduction is minimal at temperatures from 600-800°F (316-427°C), these materials are sometimes used to 800°F (427°C) where stresses are generally compressive, and there is no impact loading.
- Cold worked 300-series stainless steels lose their cold-worked effects above 800°F (427°C).
- Martensitic stainless steels (400-series) that are used in either the as-quenched condition or are tempered at low temperatures (less than 800°F (427°C) will lose their hardness if used at temperatures above 800°F (427°C). In addition, they can suffer embrittlement if used or tempered at temperatures in the 885-1025°F (475-550°C) range. Therefore, it is recommended that the 400-series stainless steel materials be tempered at 1100°F (593°C) minimum if operating temperatures will exceed 800°F (427°C).
- Duplex stainless steels embrittle due to the formation of sigma-phase at temperature above 550°F (288°C)).

Effects of Elevated Temperature on Yield Strength

Yield strength in metal alloys is a strong function of defects in their crystalline structure. These defects are formed purposely through alloying, heat treatment, cold working, etc., to strengthen materials. Elevated temperatures decrease the effectiveness of these mechanisms, effectively lowering the yield strength. Each material has its own yield strength vs. temperature profile which is dependent upon composition and material condition.

Creep at Elevated Temperature

At highly elevated temperatures, a phenomenon called creep comes into play. Creep involves inelastic behavior (that is stress is not proportional to strain) , wherein a material under a constant stress continuously deforms rather than maintaining a constant strain. The strain will slowly increase with time (hence the name "creep"). In some applications, creep becomes a significant factor in the design of a workable control valve. The temperature required to cause creep is dependent upon material composition and material condition. Creep information is usually presented in graphical or tabular form displaying the stress to cause a certain amount of permanent deformation as a function of temperature. At temperatures where creep is active, yield strength becomes irrelevant.

Effects of Elevated Temperature on Elastic Modulus

The elastic modulus decreases with increasing temperature, which means that the material becomes less "stiff". This can affect a number of components in control valves. For example, assume a bonnet bolt is torqued to provide a particular load. This load actually corresponds to a given amount of strain in the bolt at the torque limit. The valve is subsequently placed into service at an elevated temperature, which causes a reduction in the elastic modulus of the bolt material. Since the strain remains constant (assuming that all parts in the assembly have the same thermal expansion coefficients), the stress in the bolt (and thus the load) is reduced by the same proportion. Each material has its own elastic modulus vs. temperature profile which can be used to help optimize material selection for control valve components.

Coefficient of Thermal Expansion

When metallic materials are heated (or cooled), they expand (or contract) in a predictable and repeatable manner. Each alloy has its own characteristic thermal expansion vs. temperature curve which can be used to predict its dimensional change as it is heated. In general, related materials have similar thermal expansion properties, and can be grouped for general discussion purposes. The carbon steels, alloy steels, and 400 series stainless steels have fairly low thermal expansion coefficients, whereas the 300 series stainless steels have very high expansion rates. The nickel alloys fall in between.

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When selecting materials for a valve that will be used at cryogenic or elevated temperature, thermal expansion differences must be taken into account. Differential thermal expansion between plugs and cages can cause binding or excessive looseness at operating temperature. Likewise, differential thermal expansion in a body-bonnet-cage-seat ring system can cause loss of gasket load, resulting in leakage. Differences in thermal expansion rates must be either eliminated (by selection of like materials) or accounted for (by proper dimensioning of parts) when a valve is to be used at temperatures significantly different than ambient. In some cases, differential temperatures between mating parts must also be taken into account, especially in applications that involve large operating temperature gradients.

Effects of Low Temperature on Toughness

Some materials, most predominantly non-austenitic steels such as carbon-, alloy-, and martensitic stainless steels, display reduced toughness at low temperatures. If impact tests, such as Charpy or Izod, are run at a variety of temperatures on a given material of this type, an S-shaped curve results. The curve includes a lower "shelf" energy at low temperatures and an upper "shelf" energy at elevated temperatures, with a steeply sloped transition centered around the ductile-to-brittle transition temperature (DBTT). When a steel of this type is to be used at low temperature, it is customary to specify impact testing at the minimum service temperature (or a standard temperature that is even lower) to show that the material has been properly processed to meet standard minimum impact energy values.

Austenitic steels, copper alloys, and nickel alloys, and some other alloy families do not generally display a ductile-to-brittle transition due to their crystal structures. These materials are generally utilized for cryogenic service applications.

Specific Material/Environment Considerations and Limitations

The paragraphs above address some of the general phenomena which occur in various environmental situations among the various material families. The number of specific material / environment compatibility and temperature effect issues which must be addressed in the selection of control valve materials is much too large to be addressed in this chapter. However, some of the commonly encountered material / environment considerations and limitations which must be recognized follow, grouped by material type:

Gray cast iron and many of its variations:

- Lack of ductility and sensitivity to thermal and mechanical shock.

Carbon and Alloy Steels:

- The need for impact toughness verification for low-temperature applications.
- The possibility of embrittlement in carbon steels in contact with alkaline or strong caustic fluids.
- The possible conversion of carbides to graphite (sometimes called "graphitization") in carbon steel, carbon-manganese steel, nickel steel, manganese-vanadium steel, and carbon-silicon steel when exposed for long times at temperatures exceeding 800°F (427°C), and in carbon-molybdenum steel, manganese-molybdenum-vanadium steel, and chromium-vanadium steel when exposed for long times at temperatures exceeding 875°F (468°C), resulting in reduced strength and ductility.
- The potential for hydrogen blistering and / or hydrogen attack due to hydrogen exposure at elevated temperatures (above 400°F (204°C)), which causes deterioration of strength and ductility.

The possibility of stress corrosion cracking and / or hydrogen embrittlement due to exposure to cyanides, acids, acid salts, or wet hydrogen sulfide, the latter generally called sulfide stress cracking (SSC). Susceptibility is increased substantially at hardness levels greater than 22 HRC. See NACE Standards MR0175 and RP0472 for more information.

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Stainless Steels:

- Susceptibility of the austenitic stainless steels to stress corrosion cracking in chlorides and other halides (fluorides, bromides, iodides). This sometimes occurs as a result of improper . application of insulation.
- The potential for intergranular corrosion of austenitic stainless steels after being sensitized by exposure to temperatures in the range from 800-1600°F (427-871°C). A related phenomenon is intergranular stress corrosion cracking of sensitized austenitic stainless steels exposed to polythionic acid. Polythionic acid often forms when water and sulfur-containing hydrocarbons are cooled to room temperature during equipment shutdown.
- The possibility of intergranular attack of austenitic stainless steels by liquid metals, including zinc, aluminum, cadmium, tin, lead, and bismuth).
- The possibility of sulfide stress cracking in strain-hardened austenitic stainless steels, hardened martensitic stainless steels, and precipitation-hardened stainless steels. See NACE Standard MR0175 for more information.
- Embrittlement of 400-series martensitic stainless steels previously tempered at temperatures below 1100°F (593°C) when exposed to temperatures in the 885-1025°F (475-550°C) temperature range.
- Embrittlement in duplex stainless steels due to precipitation of σ-phase (sigma) and / or α′-phase during long-term exposure to elevated temperatures. The actual maximum temperature limit imposed by the ASME Boiler and Pressure Vessel Code varies depending upon the alloy, but ranges from 500-650°F (249-343°C). Short-term exposure to temperatures in the 1100-1700°F (593-927°C) range can also produce embrittlement by the same mechanism.

Nickel Alloys:

- The potential for grain boundary attack of pure nickel and chromium-free nickel alloys when exposed to sulfur at temperatures above 600°F (316°C).
- The possibility of grain boundary attack of nickel-chromium alloys above 1100°F (593°C) in reducing conditions and above 1400°F (760°C) under oxidizing conditions.
- The susceptibility of nickel-copper alloys to stress corrosion cracking in hydrofluoric acid vapors in the presence of air.

Copper Alloys:

- The potential for dezincification of copper-zinc materials.
- The susceptibility of copper alloys to stress corrosion cracking in the presence of ammonia or ammonia compounds.

Miscellaneous:

- The compatibility of packing, O-rings, gaskets, and other non-metallic parts with the process fluid. These parts are often overlooked when specifying materials of construction.
- The compatibility of any adhesives, solders, and brazing compounds with the process fluid. The presence of these materials is often overlooked when specifying materials of construction.
- The compatibility of lubricants and sealants with the process fluid.
- The effects of unusual circumstances on the service temperature. Examples include:
  - the effects of low external temperatures creating the need for impact-tested material,
  - the cooling effects due to pressure drop in the process fluid creating the need for impact-tested material,

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- the high-temperature effects of fire on low-melting point materials in valves and actuators in certain services, and
- the rapid quenching effects of fire- fighting measures, which could render some materials brittle and subject to catastrophic failure.

Many of the above considerations are mentioned in the ASME (American Society of Mechanical Engineers) Boiler and Pressure Vessel Code, Section II Part D, in ASME B31.1 *Power Piping Code*, and / or in ASME B31.3 *Chemical Plant and Petroleum Refinery Piping Code*, and as such have been recognized by industry experts as potential problems in process control equipment. As stated above, this is by no means an exhaustive listing of all environment / material interactions, but does include the most commonly encountered phenomena which must be evaluated.

Materials of Construction

This section describes the materials commonly used for various components in control valves.

Valve Bodies and Bonnets

Materials for valve bodies and bonnets must meet a number of requirements:

- They must lend themselves to manufacture of the irregular shapes that bodies and bonnets tend to have.
- They must be reliable materials with known strength properties, adequate toughness and should be produced and sold under adequate codes and standards to ensure their integrity.
- They must have adequate mechanical properties while at operating temperature.
- They must be resistant to corrosion, oxidation, and other adverse effects in the environment where they will be utilized so they will retain their integrity.

Codes for Pressure Boundary Parts

ANSI/ASME B16.34, *Valves - Flanged, Threaded, and Welding End*:

ANSI / ASME B16.34 Valves-Flanged, Threaded and Welding End (Formerly ANSI) is the basic standard used for control valves. The materials that may be used to construct the pressure containing portions of the valve are listed within the standard.

Although the materials are listed with American Society for Testing and Materials (ASTM) specifications, the corresponding American Society of Mechanical Engineers (ASME) specifications may be used interchangeably. All materials listed in B16.34 have a corresponding ASME version. (Guy Borden - what other ASME codes?)

B16.34 provides pressure-temperature ratings for a large number of common valve materials. B16.34 Annex F provides a method for determining the pressure and temperature ratings of code-approved and non-code-approved materials.

ASME Boiler and Pressure Vessel Code, Section VIII:

The only control products normally sold to Section VIII of the ASME Boiler and Pressure Vessel (B&PV) Code are silencers and some level controllers. Although control valves are seldom sold to Section VIII or to the ANSI / ASME Piping Codes (B31.1, B31.3, or B31.5), it is sound practice to use many of these requirements as design guidelines. Some of the guidelines include allowable stresses, temperature limits, welding requirements and design equations.

ANSI/ASME B31.1, *Power Piping Code*, ANSI/ASME B31.3, *Chemical Plant and Petroleum Refinery Piping Code*, and ANSI/ASME B31.5, *Refrigeration Piping Code*:

Occasionally, control valves are manufactured to these codes. An ANSI / ASME B16.34 control valve will usually satisfy the requirements. In the case of B31.1 and B31.3, fabrication welds must be performed by ASME Boiler and Pressure Vessel Code Section IX qualified welders and welding procedures, and may

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require more rigorous non-destructive testing, such as x-ray examination. Whereas B31.1 only allows the use of materials listed within B31.1, both B31.3 and B31.5 allow the use of unlisted materials provided they conform to a published specification covering composition, mechanical properties, method and process of manufacture, heat treatment, and quality control.

ANSI/ASME B16.1, *Cast Iron Pipe Flanges and Flanged Fittings* and ASME B16.42, *Ductile Iron Pipe Flanges and Flanged Fittings*:

ANSI/ASME B16.1 is the basic standard used for gray cast iron control valves, and ANSI/ASME B16.42 is the basic standard used for ductile iron control valves. Although these standards were written for gray cast iron and ductile iron pipe flanges and flanged fittings, respectively, they are the most applicable standards available for gray and ductile cast iron valve bodies and bonnets, which are not covered in ANSI/ASME B16.34. The only materials listed in B16.1 are ASTM A126 grades A and B gray cast iron. B16.42 only lists ASTM A395 grade 60-40-18.

Standard Material Specifications for Pressure Boundary Parts

The two groups of metallic material specifications predominantly used for control valves are ASTM and ASME. ASTM material specifications are prepared by ASTM committees and are generally designated as ASTM A**XXX** (for ferrous materials) . or ASTM B**XXX** (for nonferrous materials). One notable exception is ASTM A494, which covers nickel-alloy castings. ASME material specifications are prepared by the Boiler and & Pressure Vessel Committee of ASME and are designated ASME SA**XXX** (for ferrous materials) and ASME SB**XXX** (for nonferrous materials).

All ASME specifications are based on ASTM specifications; that is, if there is an ASME specification (e.g., SA216), there will be a corresponding ASTM specification (e.g. A216). The reverse is not always true. The ASME version may differ by not including all the materials in the ASTM version. The ASME version also will usually require welding per ASME Section 9 requirements, whereas the ASTM version will usually reference ASTM A488 instead. The ASME version may also have slightly different requirements (e.g., it may require certification). The subtitle will indicate any differences.

All ASME material specifications are found in Section II (Material Specifications) of the ASME Boiler & Pressure Vessel Code. Section II is divided into four parts: Part A for ferrous, Part B for nonferrous, Part C for welding materials, and Part D for allowable stress values. The temperature limits and allowable stresses listed in the code are based on metallurgical limitations of the material and on available data on mechanical properties vs. temperature. Temperature limits established because of metallurgical limitations cannot be extended. Limits established due to lack of data may be extended if the appropriate information is provided to the code committee.

Some of the commonly specified pressure retaining materials in various forms are listed in table 4. Although the ASTM and ASME standards are widely recognized, standards and codes from other standards organizations are sometimes referenced in the control valve industry. Examples include DIN (Deutsches Institut für Normung e.V., or German Institute for Standardization), JIS (Japanese Industrial Standard), ISO (International Organization for Standardization), BS (British Standards), and CEN (European Committee for Standardization). In many instances, materials included in standards issued by these organizations are similar to ASTM or ASME materials. However, in most instances, chemistry and mechanical properties overlap, but do not directly coincide with, the nearest ASTM or ASME equivalent, so some engineering judgment must be involved in the selection of alternate materials.

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Materials for Pressure Retaining Parts

Bodies and Bonnets

Valve body and bonnet materials are generally selected to roughly match the material of the mating piping. However, since the fluid velocities in valves generally exceed those in the adjacent piping, more erosion-corrosion resistant materials are sometimes utilized.

It is common practice to use either ASME material specifications or their ASTM equivalents for all pressure retaining parts, except when an end user needs a non-code material for a corrosive application where none of the code-approved materials will suffice. A non-code material is one that is not listed in any ASME Code, although it is generally intended to mean a material that is not listed in B16.34. When non-code materials are utilized for pressure retaining parts, sound engineering practices must be used to determine minimum and maximum allowable temperatures and pressure ratings.

Pressure retaining parts for a valve normally include the valve body, bonnet and body-to-bonnet bolting. Some valves may include other parts that are defined as pressure retaining. A body-bonnet spacer and the disc for a single flanged (lug type) rotary valve used for dead end service are two examples.

Gray Cast Iron

ASTM A126 Grades A and B gray cast irons are utilized for control valves for low pressure services where the lack of toughness can be tolerated. Some of the codes include restrictions on the types of services where gray iron can be utilized. When gray iron is used for flanged valves, care must be exercised when tightening flange bolts to avoid excessive bending stresses. Pressure-temperature ratings for gray cast iron are listed in table 5.

Ductile Cast Iron

ASTM A395 ductile cast iron is utilized when a stronger, more robust material than gray iron is desired, but the strength and toughness of carbon steel are not required. Some of the codes include restrictions on the types of services where ductile cast iron valve bodies and bonnets can be utilized. Pressure-temperature ratings for ductile cast iron are listed in table 6.

Carbon Steels

Carbon steel is used for a large majority of control valve applications due to its low cost and reliable performance in general applications. ASTM A216 Grades WCC and WCB are the standard materials for cast carbon steels valves. Many valve suppliers are switching to WCC from WCB (which has been the standard cast steel material for many years) due to the ASME B16.34 pressure-temperature rating advantages of WCC over WCB and the increasing popularity of WCC among both customers and suppliers. Forgings, plate and bar may also be used when certain Code restrictions are met. The various ASME codes do not allow flanges or flanged fittings (such as bonnets) to be made from hot-rolled or cold-rolled bar stock due to the unfavorable grain orientation, although they do allow the use of forged bar provided some extra non-destructive examination (liquid penetrant or magnetic particle) is performed. Only castings and forgings are acceptable for hubbed flanges and flanged components such as separable flanges and bonnets. Blind flanges may be made from castings, forgings, or plate material. Pressure-temperature ratings for some of the carbon steels are listed in table 7.

Carbon steels become relatively brittle at low temperatures, so the Codes limit their use to -20°F (-29°C). Carbon steels undergo a process called graphitization at elevated temperatures, so their use is limited to 800°F (427°C) maximum. With low temperature impact testing, the same basic materials are available as ASTM A352 Grades LCB and LCC for use to -50°F (-46°C). As is the case with WCB and WCC, LCB has historically been the standard low-temperature carbon steel material, but LCC has become the standard for new designs due to its higher strength and higher pressure-temperature limits. The upper temperature limits for LCB and LCC are 650°F (343°C) and 700°F (371°C) respectively, because these grades are generally quenched and tempered to ensure their impact resistance.

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Alloy Steels

When higher temperatures and/or pressures are involved, alloy steels are often specified for bodies and bonnets. There are a large number of alloy steel materials which valve manufacturers have supplied over the years for these applications. Most are steels with chromium and/or molybdenum added to enhance their strength and resistance to tempering and graphitization at elevated temperatures. The molybdenum and chromium additions also increase their resistance to erosion/corrosion in flashing applications such as heater drains. The most popular material is ASTM A217 grade WC9. Forgings, plate and bar may also be used when certain Code restrictions are met.

In the past, ASTM A217 grade C5 (5% Cr, ½% Mo) was commonly specified for applications requiring chromium-molybdenum steel castings. However, this material is difficult to cast, and tends to form cracks when welded. If casting defects are encountered during machining, weld repair is very difficult, and bodies must sometimes be scrapped and re-ordered due to proliferation of cracking. For this reason, suppliers are standardizing on WC9 (2¼% Cr, 1% Mo) as the standard chromium-molybdenum steel casting. WC9 is preferred by the foundries, and is much easier to machine and weld than C5. Experience has shown that WC9 and C5 have essentially equivalent resistance to flashing damage; CF8M and 316 are even better. Pressure-temperature ratings for some of the alloy steels are listed in table 8.

For temperatures below -50°F (-46°C), low alloy steels, some with 1% to 9% nickel, are available. These steels are impact tested for service at temperatures as low as -175°F (-115°C), and are typically only available by special order. Pressure-temperature ratings for some of the nickel steels are listed in table 7. Austenitic stainless steels are sometimes utilized because of their ready availability and their acceptability for use at very low temperatures without impact testing.

Ferritic Stainless Steels

Ferritic stainless steels are seldom used for control valve bodies and bonnets. The major reason is that the ferritic stainless steels with the most attractive properties cannot be cast. ANSI/ASME B16.34, does not list any ferritic stainless steel materials. The ASME B&PV Code does list allowable stress values for 405, 430, 26-4-2, 27-1, 29-4 and 29-4-2. Other codes may list some of these materials. Ferritic stainless steels may be used for valves produced from wrought material such as plate butterfly bodies or fabricated angle style bodies. The minimum temperature is -20°F (-29°C) for all ferritic stainless steels. The maximum temperatures range from 600°F (316°C) to 1200°F (649°C).

Martensitic Stainless Steels

Martensitic stainless steels are not widely used for control valve bodies and bonnets. Their primary use is for wellhead and refinery applications. ANSI/ASME B16.34does not list any martensitic stainless steels. The ASME B&PV Code and some other Codes do list allowable stress values. The maximum temperatures vary by alloy but all are limited to a minimum temperature of -20°F (-29°C).

Type 410 stainless steel bar, pipe and tubing, ASTM A182 Grade F6a forgings and the ASTM A217 grade CA15 are rated to 1200°F (649°C), but the long-term usable strength above 900°F (482°C) is very low. Some suppliers recommend limiting its use to 800°F (427°C) maximum.

Most CA15 castings have been replaced by a newer grade of material CA6NM. This is a modified martensitic stainless steel which has improved casting properties and superior corrosion resistance and toughness. CA6NM has a slightly lower carbon content and increased nickel and molybdenum. The forged version, F6NM, is not listed in the Codes. These materials are purchased in the quenched and tempered or normalized and tempered condition. CA6NM is generally limited to a maximum temperature of 800°F (427°C).

Austenitic Stainless Steels

The conventional austenitic stainless steels are basically the 300 series alloys. The control valve industry standard for stainless steel bodies and bonnets is ASTM A351 grade CF8M (the cast version of 316). With its

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nominal 19% Cr, 10% Ni, 2% Mo composition, CF8M is a relatively low-cost material with excellent low and high temperature properties and excellent resistance to corrosion in a wide variety of environments. Type 316 forgings, plate, pipe are also used.
For temperatures below -50°F (-46°C), austenitic stainless steels should be used. Other carbon and low alloy steels are permitted by ANSI, however, long leadtimes and low volumes make the austenitic stainless steels more practical. CF8M and 316 can be used to temperatures as low as -325°F (-198°C) without impact test requirements. CF8 and 304 can be used to temperatures as low as -425°F (-255°C).

CF8M's high chromium and molybdenum contents give it even better resistance to erosion in flashing applications than WC9 or C5 material. CF8M and the other austenitic stainless steels are also used in many applications for their high-temperature pressure ratings. However, because of their relatively high thermal expansion rates, the austenitic stainless steels are more susceptible to thermal fatigue than carbon and alloy steels in applications involving high-temperature thermal cycling.

304L, and its cast form CF3, are the standard materials for nitric acid service. 317 and CG8M are preferred materials for the pulp and paper industry. The increased alloy content compared to 316 and CF8M provides the additional corrosion resistance required in many chloride-containing environments. Use of any conventional austenitic stainless steels in chloride containing environments must be limited to 160°F (71°C) maximum to prevent chloride stress corrosion cracking (SCC). Even lower maximum temperatures should be observed at low pH levels.

347 and CF8C are generally limited to special applications where a stabilized grade is required to prevent sensitization, such as in high-temperature, sulfur-containing hydrocarbon services. The columbium (niobium) content also provides slightly higher pressure ratings at elevated temperatures compared with 316/CF8M. There is no improvement in corrosion or other properties compared to 316 and CF8M.

Pressure-temperature ratings for some of the austenitic stainless steels are listed in table 9.

Low Carbon Grades: Low-carbon versions of the austenitic stainless steels (such as 316L and its cast equivalent, CF3M) contain a reduced carbon content (generally 0.03% maximum) in order to avoid sensitization of the heat-affected zone during welding. The use of these grades is justified for buttwelding end bodies that will be used in corrosive applications, since the installation welds will be full-penetration and post-weld solution heat treatment is not possible. For flanged bodies, industry experience has shown that the low-carbon grades of the austenitic stainless steels are seldom required. The slight sensitization which occurs on minor weld repairs of casting defects only creates problems in applications which produce significant corrosion on the material. Low-heat-input weld procedures and L-grade weld filler materials minimize sensitization concerns even further. Major repairs should be performed by the foundry before the solution heat treatment process. High Temperature Grades:

For high temperature applications, the ASME Boiler and Pressure Vessel Code requires that the carbon contents for CF8, CF8M, and CF8C be in the upper half of the carbon range, or 0.04 to 0.08% for service at temperatures greater than 1000°F (538°C). The H grades are specified for the wrought forms of 304, 316, 347, etc. The H grades have carbon contents of 0.04 to 0.10%.

Super-Austenitic Stainless Steels

ANSI/ASME B16.***34*** does not list any of the newer super-austenitic stainless steels; however, several of the older alloys are listed. The ASME Boiler and Pressure Vessel Code does list allowable stress values for a number of the newest super-austenitic alloys, several of which are finding increased usage as control valve body materials. UNS S31254 (Avesta 254 SMO®) is the most widely used of the super-austenitic grades that are sometimes referred to as "6 Mo" materials due to their minimum molybdenum content of 6%. Castings are supplied to ASTM A351 grade CK3MCuN. Castings in CK3MCuN may required additional specifications to

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ensure that the castings will have adequate integrity, weldability, and corrosion resistance. Wrought products are purchased under regular ASTM / ASME specifications as UNS S31254.

One of the older alloys which could be classified as a super-austenitic is N08020 (Carpenter 20Cb-3®), commonly called alloy 20. ANSI / ASME B16.34 includes this alloy as castings per ASTM A351 grade CN7M. Its use is declining somewhat with the advent of the newer super-austenitics.

Duplex Stainless Steels

Duplex stainless steels are generally defined as stainless steels containing approximately 40-60% austenite and 60-40% ferrite. The duplex stainless steel materials offer better resistance to crevice corrosion and pitting in chloride-containing environments than the conventional austenitic stainless steels, at costs lower than those of the super-austenitic materials. These materials are commonly utilized for seawater applications, and are sometimes even used to prevent external corrosion of valves that are exposed to salt-spray. ANSI / ASME B16.34 does not list any duplex stainless steels. The ASME Boiler and Pressure Vessel Code does list allowable stress values for CD4MCu, S32550 (wrought Ferralium®255), S31803 (wrought 2205), wrought S32404 (Uranus® 50) and S32750 (wrought SAF® 2507). Note that CD4MCu is the only cast duplex SST listed in the ASME Boiler and Pressure Vessel Code. Many control valve producers supply cast 2205 (ASTM A890 grade 4A or CD3MN) and cast Ferralium® 255 (CD7MCuN). These grades are non-code approved and must be producer rated.

Due to the formation of σ-phase at elevated temperatures, duplex stainless steels are limited to a maximum service temperature of 500 to 600°F (260 to 316°C). The formation of σ-phase adversely affects both the toughness and corrosion resistance of the material. Welding of duplex alloys can also be somewhat difficult due to the potential for forming σ-phase upon cooling.

At this point in time none of the super duplex stainless steels are listed in any of the Codes. Activities are now under way to add alloys such as S32760 (Zeron 100®) to the Codes.

Nickel Alloys

The nickel base alloys listed in ANIS / ASME B16.34 are N02200 and N02201 (Nickel 200 and 201), N04400 and N04405 (Monel® 400 and 405), N06600 and N06625 (Inconel® 600 and 625), N10001, N10665, N10002, N06455, N10276, N10003, and N06002 (Hastelloy® B, B2, C, C4 and C276, N and X). The only cast versions of these alloys listed are N12MV (Hastelloy® B) and CW12MW (Hastelloy® C). Both N12MV and CW12MW have been replaced by newer casting alloys with superior castability, corrosion resistance and weldability, etc., including alloys such as CW2M, CW6M, and N7M. Due to their high costs, the nickel alloys are generally only used for severely corrosive environments that cannot be handled by stainless steels.

Titanium, Zirconium, and Tantalum

ANSI / ASME B16.34 does not list any of the refractory metals. However, the ASME Boiler and Pressure Vessel Code lists titanium and zirconium. The Code lists 3 wrought and 2 cast grades of titanium, all of which are commonly used for control valves. The ASME Boiler and Pressure Vessel Code lists 2 wrought zirconium grades. Cast zirconium is not listed in the Code, so the supplier must use ASME procedures to determine ratings.

Because of their reactivity with oxygen and nitrogen, the refractory alloys are difficult to produce, especially in the cast form. Titanium is more expensive than the nickel alloys, and is generally used for very aggressive chloride-containing environments that cannot be handled by the stainless steels or nickel alloys. Zirconium is even more expensive than titanium. It is also used for some very severely corrosive environments.

Tantalum is a very inert material that is resistant to many environments that cannot be handled by any other material. Unfortunately, it is very expensive, difficult to produce, and has very poor mechanical strength.

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Therefore, when it is utilized it is usually supplied as a liner in a steel body. In these cases, the body is generally rated based upon the structural body material.

Copper Alloys

ANSI/ASME B16.34 does not list any copper base alloys. However, the ASME B&PV Code and some other Codes do list several alloys. The maximum temperature varies by alloy but all are limited to a minimum temperature of -325°F (-198°C).

Bolting

There are many different grades of bolting materials, and their grade designations are different than those used for other products. While there are a large number of materials with slightly different compositions, heat treatments, and resulting mechanical characteristics, most needs can be met with just a few grades of material.

The most common bolting materials used in control valves are ASTM A193 grades B7, B7M, B8M, and B16. Corresponding nut materials are listed in ASTM A194. The most commonly used nut materials are grades 2H, 2HM, and 7. Bolts for low-temperature applications are covered in ASTM A320. The grades in A320 are similar to, and in some cases identical to, those in A193. The ASME B&PV Code allowable stresses (required per ANSI/ASME B16.34) are the main criteria used to determine bolting materials for most applications.

Grades B7 and L7 Bolts and Grades 2H and 7 Nuts

ASTM A193 grade B7 bolting is actually an AISI 4140 or similar chromium-molybdenum alloy steel which has been heat treated to provide certain mechanical properties. Grade B7 is the standard bolting material supplied in the vast majority of control valves, offering excellent strength over a large temperature range, thermal expansion rates closely matching those of WCB, WCC, and WC9, excellent availability, and reasonable cost. Grade B7 can be used from -50°F to 1000°F (-46°C to 538°C), although above 700°F (371°C) its allowable stresses are lower than those for grade B16. B7 bolting is generally used in conjunction with ASTM A194 grade 2H nuts, which are quenched and tempered medium-carbon steel. In some cases, A194 grade 7 nuts, which are chemically and mechanically equivalent to B7 studs.
ASTM A320 grade L7 bolting is actually grade B7 which has been impact tested to demonstrate toughness to -150°F (-101°C). L7 bolting is generally used with grade 7 nuts that have been impact tested at -150°F (-101°C).

Grade B16

ASTM A193 grade B16 is a modified G41400 material, with additions of vanadium and extra molybdenum to give it superior high temperature properties. It also matches the thermal expansion properties of WCB, WCC, and WC9. It is mainly used for temperatures above 700°F in conjunction with alloy steel bodies and bonnets. It is generally used with grade 7 nuts.

Grade B8M

ASTM A193 grade B8M bolting is S31600 stainless steel. B8M is used for high- or low-temperature applications or to match the thermal expansion characteristics of a CF8M body and bonnet. B8M is available in two strength levels, Class 1 and Class 2. B8M Class 1 is manufactured from annealed bar stock, whereas B8M Class 2 is manufactured from strain-hardened bar stock. Allowable stress levels are listed for B8M Class 1 bolting up to 1500°F, and above 1000°F its allowable stress values are greater than for B16. B8M Class 2 bolting has higher allowable stress values up to 800°F due to the strain-hardening. However, above 800°F, the strain-hardening effects are reduced by temperature effects. For this reason, the allowable stresses for B8M Class 2 are equal to those for B8M Class 1 from 850°F to 1000°F, and its use is not permitted above 1000°F. The corresponding nut grade is ASTM A194 grade 8M, which is only available in the annealed condition. Standard ASTM A193 grade B8M studs and A194 grade 8M nuts can be used to -325°F (-198°C) without impact testing.

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ASTM A193 also includes a 316 stainless steel grade which is solution annealed after all threading and forming operations. This grade is designated B8MA Class 1A, and should be used whenever type 316 bolting is required for NACE MR0175-compliant constructions.

For applications involving temperatures below -325°F (-198°C), ASTM A320 provides the option of impact testing B8M bolts and corresponding ASTM A194 grade 8M nuts down to -425°F (-254°C).

Coatings

The use of internal and external coatings and plating are not addressed by the Codes. They are generally used to prevent corrosion on carbon and low alloy steels or wear from abrasive fluids on any material. Several commonly used coatings are described below.

Electroless Nickel Coating (ENC)

Electroless nickel coating can be used to protect steel bodies and bonnets from corrosion. Common uses are in sea water, sour gas, and oil. Typical ENC coating thickness would be 0.010" (0.25 mm). For all practical purposes, ENC will contain some pin-holes like all coatings do. Only through very extensive inspection and testing can one be reasonably comfortable that ENC is pin-hole free. Once in service, however, the ENC may become worn, mechanically damaged or suffer chemical attack, exposing the base metal.

Aluminizing

Aluminizing is a high temperature, gaseous diffusion process for protecting steel and stainless steel from high temperature corrosion. The aluminum-containing compound layer formed on the surface is particularly resistant to sulfide attack. Aluminized steel is a very economical material for refineries where sulfide attack is a common problem. Aluminized ferritic and austenitic stainless steels have excellent resistance to carburization.

Boronizing

Boride diffusion coatings are used to prevent erosion of internal valve surfaces; trim parts and / or bodies. Several different compounds are formed depending on the base metal and the presence of other species in the furnace atmosphere. Typical compounds include boron carbides, nitrides and silicides and chromium and titanium borides. Thicknesses are generally less than 0.0005" (0.01 mm) on austenitic stainless steels and near 0.010" (0.25 mm) on steels and martensitic stainless steels. Application of boronizing on stainless steels may be limited by the adverse effects of the process on the base material's corrosion resistance.

Sprayed Coatings

Plasma, flame sprayed, and high-velocity oxy-fuel (HVOF) coatings can be applied to improve wear resistance, and in some instances, corrosion resistance. However, due to the nature of spray processes, there are limitations regarding coating of internal diameters and complex internal and external geometries. For this reason, the spray processes are generally used for coating the bores of butterfly and ball valve bodies and the wear surfaces of trim parts in various valve styles. Coating materials include chromium oxide, tungsten carbide, chromium carbide, cobalt-chromium-tungsten (Stellite®) alloys, cobalt-chromium-molybdenum-silicon or nickel-chromium-molybdenum-silicon (Tribaloy®) alloys, nickel-chromium-boron (Colmonoy®) alloys, and many other wear resistant materials. Unlike weld overlay methods, which are metallurgically bonded to the base material, spray coatings are attached by mechanical bonding. Under impact or localized loading conditions, sprayed coatings are subject to failure by spalling. Furthermore, although the spray processes can be used to apply corrosion-resistant alloys, the corrosion resistance of sprayed coatings does not match that of weld-overlays due to oxidation of the alloy powder during application. In addition, the coatings always contain some degree of porosity which renders them ineffective for protecting non-resistant base materials.

Polymeric Liners

The high cost of chemical resistant alloy valves has created a niche for special linings in low cost steel valves which resist chemicals. These constructions are essentially composites which employ the structural strength of steel to

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retain process pressure while the lining provides a protective corrosion barrier for the valve body. Butterfly, ball, gate, plug and globe valves are all made with plastic liners. The most chemical resistant of lining materials are the fluoropolymers, PTFE (polytetrafluoroethylene), FEP (fluorinated ethylene-propylene), PFA (perfluoro alkoxy alkane) and PVDF (polyvinylidiene fluoride). These linings have almost universal chemical resistance with the fully fluorinated polymers (PTFE, FEP & PFA) having a definite edge over PVDF. A variety of molding processes such as rotomolding, injection or compression molding can be used depending on which polymer is to be molded and the configuration of the valve body. Linings tend to be thick to reduce the permeation rate of the process through the lining to interact with the valve body wall. Some designs use the liner as the actual sealing surface, while others employ additional seal components that mechanically join to the liner.

Thermosetting rubbers are also used as liners for valves, especially butterfly and pinch valves. Rubber liners are typically compression molded into the steel or cast iron valve body. The liners are used for some combination of chemical compatibility, abrasion resistance and tight shut-off. A variety of rubber materials are employed depending upon the process. Hydrocarbon resisting elastomers such as nitrile and fluoroelastomer are used for liquid and gaseous fuels. Water and steam resistant elastomers would include ethylene-propylene and tetrafluoroethylene/propylene copolymer rubbers. Extremely abrasion resistant rubbers such as polyurethane and natural rubber are applied to slurry applications where high solids content can erode through steel valves in just hours. The compliant rubber lining tends to absorb impact energy and provide wear life magnitudes longer in duration than metal.

Some steel valves are coated internally with special organic coatings that are spray or brush applied. These coatings are usually epoxy or phenolic based and are intended to impart additional barrier resistance to mildly corrosive environments such as sea water.

Trim Material Selection

The heart of any control valve is the trim set. If the trim fails to perform properly for any reason, the valve will no longer be able to . properly control the process. A number of factors must be considered to insure that trim materials will perform as required. These factors fall primarily into two categories:

1. The environmental compatibility of the materials, including general corrosion resistance and resistance to environmentally assisted cracking, etc.
2. The mechanical suitability of the materials, including strength and wear resistance.

To make matters difficult, these categories conflict in many instances, making it difficult or impossible to satisfy all considerations with a single material. In these cases, the best compromise must be identified.

Environmental Considerations

Corrosion is the first item that should be reviewed in the selection of trim materials. All of the forms of corrosion, such as general corrosion, localized corrosion (pitting and crevice), stress corrosion cracking, etc., should be considered. Next, other environmental factors such as temperature restrictions, should be reviewed. A listing of materials with acceptable environmental compatibility should result from this review. Based upon this list, it may be necessary to restrict the remainder of the valve selection process to particular valve and / or trim styles, since some materials do not lend themselves well to particular designs due to limitations in mechanical properties.

Mechanical and Physical Properties

When selecting materials, the mechanical and physical properties which must be considered can vary greatly depending upon the trim component and the valve design. Obviously, the properties which are important in the selection of a plug material are different from those used in the selection of a cage material.

Materials of Construction

This section describes the essential materials considerations for various, common control valve trim components, explains the more commonly encountered problem areas in valve trims, and lists some commonly used materials.

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Globe Valve Components:

Plugs

Valve plugs provide throttling control and shutoff in globe valves, and are directly impinged by the flow stream. The seating surface on the valve plug must be capable of withstanding the seat loads required for shutoff. It must also withstand the erosive forces generated by fluid jets from drilled-hole, slotted, and other tortuous path trim, or during low-lift throttling. In cage-guided valves, the plug guide surfaces must resist galling and excessive wear when sliding against the cage material, and must withstand the erosive action caused by clearance flow between the plug and cage. In post-guided or port-guided constructions, the plug material must provide good resistance to galling in conjunction with the bushing or seat ring material, respectively.

Seat Rings

Seat rings in globe valves work with the valve plug to provide shutoff. They must be able to withstand the seat loads required for shutoff as well as the erosive forces caused by the high fluid velocities which can be encountered during low-lift throttling. In some designs, the seat ring is an integral part of the cage. In certain seat ring designs, the seat ring flange must be strong enough to withstand the bending loads imposed on it by a cage, a seat ring retainer, and/or a plug/seat load. In port-guided designs, the seat ring material must provide good galling and sliding wear resistance in conjunction with the plug material.

Cages

Valve cages can serve a number of functions in a globe valve depending upon the valve design. In most globe valves, the cage provides plug guidance and is involved in flow characterization.

In clamped seat ring designs, the cage transfers a portion of the bonnet bolt loading to the seat ring to hold it in place and maintain gasket loading. These cages are required to withstand axial compressive loading due to the bonnet bolt load, and must provide good sliding wear properties in combination with the plug material. The cage material also must be compatible with the body material from a thermal expansion standpoint to prevent alteration of gasket loading with temperature changes in the cage and the body.
In hung cage designs, the cage is held in place by an integral flange which is clamped between the body and bonnet. Many hung cages contain an integral seat. These cages must be made from a material capable of handling the tensile loading due to seating forces, as well as the compressive stress on the clamped integral flange. They can also be subject to high vibrational forces, in which case the material must be resistant to fatigue .

In cage-guided trim designs, the cage is subjected to a significant portion of the pressure drop which occurs in the valve, and as such must withstand circumferential loading caused by the pressure differential across the cage wall (tensile in flow-up applications, compressive in flow-down applications).

All cages must provide good sliding wear properties in combination with the plug material. In some applications, the cage material is required to provide good erosion resistance. This is especially true in tortuous-path designs, where there is a great deal of interaction between the fluid and the cage surfaces.

Bushings

In post-guided constructions, a bushing serves as a guide surface for the top of the plug or the stem directly above the plug. The bushing must be resistant to galling in conjunction with the plug or stem material.

Stems

The globe valve stem connects the valve plug with the valve actuator through the bonnet packing box. The valve stem must be resistant to both general corrosion and pitting so that leakage and/or damage to the packing will not occur. The stem must be strong enough to sustain the actuator loads without buckling or yielding. In certain designs, the stem must provide good sliding wear properties in combination with one or more guide bushings. Finally, the connection between the valve stem and the plug must be able to withstand

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all operational loads, including those imposed by seating and/or unseating the valve plug (especially in flow-down, unbalanced applications), changing stem-force gradients, and flow-induced vibration, without loosening and/or breaking.

Materials-Related Problem Areas in Globe Valve Trim:

Plug O.D. and Seat Line Erosion

One of the major reasons for replacement of valve trim is erosion of the plug O.D. and seat line due to impingement of flow from cage openings and/or clearance flow between the cage and the plug. This type of damage can be minimized by a number of techniques. Avoiding extensive operation of the valve at travels that result in clearance flow or cage-opening impingement is the best solution. However, this type of approach is a valve sizing and valve operation issue, and may not be possible in many applications. Special trim designs can alleviate the problem in some instances. If clearance flow or cage-opening impingement cannot be avoided, the use of erosion-corrosion resistant materials or coatings on the valve trim is recommended.

Plug/Seat Interface Erosion (Wire Drawing)

This problem, often called "wire drawing" because the damage looks like grooves caused by drawing a wire over the seat surface, occurs when the plug and seat ring surfaces become locally damaged due to cavitation, erosion, corrosion, galling, or entrapment of foreign particles. Subsequent localized flow past the seat during shut-off conditions causes linear erosion-corrosion and/or cavitation damage. The use of harder materials generally improves resistance to this type of damage, although in many cases alloys with inherent erosion-corrosion resistance are more effective. Cobalt-base alloy 6 is one of the most resistant metallic material to this type of damage, except in cases where it is particularly susceptible to erosion-corrosion, such as in hydrazine-treated boiler feedwater. Tungsten carbide and ceramics are even more resistant.

The problem can also occur when adequate materials are used, but seat loads are too low and/or seat surfaces are improperly matched. This commonly occurs when seats are over-lapped, causing a large seat area which requires excessive actuator force to produce tight shutoff. Lapping should cease when a narrow, matched seat line is attained.

Cage Opening Erosion (Wire Drawing)

Damage identical in appearance to wire drawing on plug and seat ring seating surfaces sometimes occurs at the bottom of cage openings, particularly in severe service applications. Generally, this damage is caused by fine particulate matter (such as sand, catalyst fines, pulverized weld slag, or other foreign material) which is entrained in the process fluid. This type of damage is often eliminated through the use of strainers or separators which remove the material upstream of the valve, or through the use of m If these approaches are not possible, very hard, abrasion-resistant materials may provide increased service life assuming they are adequate from a corrosion standpoint.

Cavitation Damage on Plugs, Seat Rings, and Cages

Cavitation damage is best eliminated by utilizing special trim designs which produce pressure drops in multiple stages, often involving tortuous-path technology. These anti-cavitation trims are not cavitation resistant, but instead prevent cavitation from occurring. They eliminate the need for special, expensive materials.

There are instances where the use of anti-cavitation trims is not feasible either for economic or practical reasons. In those cases, the trim materials used should be resistant to cavitation damage. Cobalt-base alloy 6 is generally considered to be the best metallic material for resistance to cavitation damage. Tungsten carbide is perhaps the ultimate cavitation-damage resistant metal-ceramic composite material. Under low-intensity

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cavitation, elastomeric materials are sometimes very resistant to damage. Refer to Chapter 7 for more information on cavitation and cavitation-resistant materials.
It must also be recognized that cavitation creates a great deal of noise and vibration. Therefore, the trim design and all materials utilized should be resistant to fatigue if cavitation is occurring.

Plug/Cage Interface Galling

As was discussed briefly above, plug and cage materials for cage-guided valves must be resistant to galling, especially in high-temperature, high-pressure applications. It can be difficult to provide cage-guided valve trim which is resistant to galling and will also withstand many commonly encountered corrosive environments. The common coatings utilized to protect the 300-series stainless steels (cobalt-alloy hardsurfacing, electroless nickel, hard chromium plating, etc.) are often not resistant to the environments which require the use of duplex and superaustenitic stainless steels, nickel alloys, titanium, and other special materials. In most instances, it is best to utilize alternate trim styles, such as post-guided constructions, which allow the use of plastic guiding surfaces for the uncoated, corrosion-resistant alloys required in these severely corrosive applications.

Port-Guided Plug/Seat Interface Galling

Port-guided trim is utilized in many applications which require precise control of fluids at low flow rates. Many of these applications involve fluids which are relatively aggressive from a corrosion standpoint, but the flow conditions are not particularly demanding. Austenitic stainless steels are often specified for both the plug and seat ring, a situation which commonly results in severe galling, sometimes to the point that the plug "welds" to the seat ring and cannot be extracted.

Hardened 400-series stainless steels (most commonly types 416 and 440C) are often utilized for both the plug and the seat ring in non-corrosive or very mildly-corrosive applications. Austenitic stainless steel seat rings are generally hardsurfaced both on the seating surface and in the port. Mating plugs are hardsurfaced on the seating surface as well as on the entire plug tip. Smaller trim sets often utilize solid alloy 6 seats and plug tips. In very erosive applications, tungsten carbide plug tips and seat ring inserts can be employed.

Post-Guided Plug/Bushing Interface Galling

Since post-guided trims are often utilized for high-alloy constructions, the problem of galling between the bushing and the plug post or stem is generally solved through the use of a plastic-lined bushing. Generally, the bushing jacket is the same material as the remainder of the trim. Bushing liners are commonly some type of plastic alloy which is predominantly PTFE, ensuring its applicability in a wide variety of environments.

In trim sets made from less corrosion-resistant materials, metal alloys with reasonable wear properties are often employed. In trim sets of hardened 400-series or 17-4 precipitation-hardenable stainless steel, 17-4 is often utilized as the guide bushing. In 300-series trim, cobalt alloy 6 is often used, and the plug post is generally hardsurfaced.

Plug/Stem Connection Failure

The plug/stem connection is one of the most misunderstood features in a globe-style control valve. When the plug/stem connection fails, it generally attracts a great deal of attention, since it is one of the few failures which will render the valve completely inoperable. However, assuming that the connection is properly designed, a plug/stem connection failure is usually the result of some other problem which initially appears to be of secondary importance. Some of the things which can lead to plug/stem connection failure include: failure of a piston ring or corrosion of the plug O.D. or cage I.D., both of which can contribute to excessive lateral motion of the plug on the stem; improper valve sizing, which can result in operation of the valve just off of the seat, with accompanying vibrations and/or flow instabilities, etc.

There are a large number of different plug/stem connection techniques which have various advantages and disadvantages from a metallurgical standpoint. Although a full-penetration-welded plug/stem assembly may

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at first glance appear to be the strongest and most reliable valve stem connection design, it forces some material compromises. For example, one of the most commonly utilized stem materials is strain-hardened 316 stainless steel. If welding is performed on cold-worked materials, the weldment, and usually a portion of the heat-affected zone (HAZ), will essentially end up being annealed material, which is much weaker and less resistant to fatigue than the cold-worked material. This weaker region is located in the portion of the stem that experiences the most stress. Also, if heat treated materials (such as types 410 or 17-4) are utilized, their properties are also usually compromised during welding, unless complete re-heat treatment will be performed on the assembly after welding, which is usually not practical.

Plug/stem assemblies are sometimes supplied with a threaded joint supplemented with a fillet weld on top of the plug. Stem connections are sometimes even modified by customers in this fashion because it is viewed as a more robust anti-rotation design (which it probably is) and as an overall strength booster. However, welding on the top of a threaded connection actually decreases the strength and robustness of the connection by relieving the tensile stress preloads which keep the threaded portion of the connection tight. This essentially results in cyclic loading of the weld fillet. Experience has shown that top-welded connections are less reliable than threaded and pinned connections.

If welding must be performed on a valve stem connection, the bottom of the stem is the best location for the weld (of course, this option is only available in balanced plug constructions where the bottom of the stem is accessible and on weldable plug and stem materials). Bottom welding provides robust anti-rotation performance, and avoids relief of the tensile stress preloads introduced during tightening of the threaded portion (provided welding heat input is kept to a minimum).

Since no welding or other heating is involved in assembly, the threaded and pinned joint can be the most cost-effective method of attaining a reliable, high-strength stem connection, provided certain design criteria are met. The most commonly expressed concern regarding this design is that it could loosen and break or simply come apart in service. However, loosening of the joint can be prevented by using the same concepts used in the design of reliable bolting systems in other cyclic loading applications, such as those used in proper valve body/bonnet bolting. The geometry and pre-loading (torqueing) procedure should be such that the stresses imposed on the joint during assembly are higher than the cyclic loads encountered in service. Whereas this is not accomplished with simple tap and die-cut or single-tooled threads, it can and has been accomplished using proprietary valve stem connection designs.

Common Globe Valve Trim Material Combinations:

The most popular trim combination in cage-guided globe valves consists of a hardened 400-series stainless steel seat ring and plug (generally type 416), a strain-hardened 300-series stem (usually type 316), and a precipitation-hardened stainless steel cage (such as 17-4). This combination will perform well in most general valve applications.

At higher temperatures, or when conditions become somewhat more corrosive, these materials are generally replaced with 300-series stainless steel (usually type 316), or one of the special 12% chromium stainless steels designed for high-temperature service. To improve galling resistance, the plug and seat ring are often hardsurfaced with cobalt-base alloy 6, and the cage is often coated with either chromium plating (useful to around 600°F (316°C)), special chromium coating (good to as high as 1100°F (593°C)), or electroless nickel (useful to approximately 650°F (343°C)), or is nitrided (useful to well over 1100°F (593°C) in non-corrosive applications). In elevated temperature applications, consideration must be given to thermal expansion rates of trim materials to avoid undesirable changes in plug/cage clearance. In valves which are post-guided, plastic-lined guide bushings are sometimes utilized along with 300-series materials without coatings where application temperature permits.

Cage-guided trims in higher alloys pose a difficult problem. The more corrosion-resistant, non-hardenable high-nickel alloys have poor resistance to galling. Precipitation-hardenable versions (such as K500, X750, 718, and 725) can be utilized for some services, but even they don't possess complete resistance to galling. The coatings which are commonly used to protect the stainless steels from galling are generally not resistant to the

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environments which dictate the use of high-nickel alloys. This is one of the main reasons that the high-nickel alloys are generally provided in post-guided valve designs with plastic-lined guide bushings. See the coatings section for a discussion of special coatings which are available to increase wear and galling resistance of titanium and zirconium alloys when used in trim applications.

Some trim designs make use of elastomeric or plastic seats for tight shutoff. One of the most prevalent materials used for this type of application is PTFE, since it has nearly universal chemical resistance. Elastomeric materials are also utilized, but care must be taken to ensure that the elastomer is resistant to chemical attack in the process environment. Several elastomer grades are generally required to cover a broad range of applications.

Rotary Valves Components:

Disk/Ball/Plug

In rotary valves, the closure member is generally either a disk, a ball, or a plug. There are a number of different variations in shapes and configurations in each category. For example, disks may be plain, relatively flat, and concentric with the shaft axis (the conventional butterfly valve), or they may be eccentric and/or cammed. Balls range from complete spheres with a through hole to segments of spherical surfaces, and are usually operated concentric to the shaft axis. The term "plug" is generally reserved for heavy-duty versions of cam-operated spherical surfaces. In all cases, these components are subject to flow erosion, and must interface with the seal to provide shutoff. Eccentric and cam-operated designs tend to only interface with the seal at low travels. Concentric designs generally contact at least a portion of the seal throughout the full travel range. In any case, the disk/ball/plug must provide adequate wear resistance in conjunction with the seal material to maintain good shutoff, while providing an appropriate level of flow-erosion resistance. Throughout the remainder of this discussion, this component will be referred to as a "ball", but the points made will also be applicable to disks and plugs.

Seal/Seat

As the name implies, the seal provides the surface against which the ball seals to provide shutoff. The seal/seat in rotary valves tends to vary more in design and material than any other trim component. However, in all cases, the seal must provide good wear resistance in conjunction with the ball in order to maintain good shutoff. In addition, since seals/seats are often designed with thin cross-sections, the material must resist both flow erosion and deformation/failure due to flow-induced forces.

Shaft

The shaft in a rotary valve transmits torque from the actuator to the ball. The shaft is required to withstand the shear and bending stresses imposed by the pressure-drop across the ball, and the torsional stresses due to actuation. In addition, it must provide good wear resistance and low friction in combination with the bearing surfaces to prevent galling, excessive torque requirements, and/or excessive shaft deflection over time.

Bearings

Most rotary control valves utilize bearings to support the shaft and provide a better wear surface than would be afforded by the valve body material. Bearings range from hardened and/or coated metal to plastic-lined metal to all plastic constructions. The bearings must provide good galling resistance and wear and low friction in conjunction with the shaft material.

Pins

Whereas some rotary control valves utilize a trunnion or spline connection between the ball and the shaft, most utilize some type of square key, taper pin, taper key, or groove-pin joint. In these designs, the pin or key must provide adequate strength to resist the shear stresses imposed by operation of the valve. In some cases, the thermal expansion coefficient must also be appraised to prevent loosening or excessive tightening during thermal excursions.

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Materials-Related Problem Areas in Rotary Valve Trim:

Ball/Shaft Connection Failure

Cyclic service conditions at high pressure-drops can cause fatigue in certain connection types, particularly in designs which involve a large gap between the bearings and the shaft bore in the ball. Materials strengthened by either strain-hardening or heat treatment are generally utilized for shafts and pins to prevent fatigue due to the bending and torsional stresses imposed.

Ball and Seal Wear

Since the seat surfaces on the seal and ball govern the shutoff capabilities of the valve, these surfaces must remain smooth and relatively defect-free. Various metallic materials are used for both parts, and the wear resistance of one or both parts is often improved by hard chromium plating, electroless nickel coating, hardsurfacing, etc. The addition of these wear-resistant layers generally provides significant improvement in shutoff after a large number of valve cycles. However, as is the case with globe valves, the material/coating systems which provide the best wear properties are generally not the ones which are resistant to a wide variety of corrosive environments. In many of these cases, a metal ball/plastic seal configuration provides the best combination of corrosion resistance and long-term shutoff, because a wear-resistant coating is not required on the ball component. In instances where corrosion resistance and a metallic seal are both desired, ball/seal material combinations customized for the particular corrosive environment are usually required.

Shaft/Bearing Wear and Galling

Wear and galling between the shaft and bearings is a difficult problem to overcome. Metallic bearings are generally made from the same types of materials used for plugs in globe valves, i.e., the hardened 400-series stainless steels, cobalt-base alloy 6, etc. Sometimes these bearings are coated with solid film lubricants, such as MoS_2 (molybdenum disulfide, or moly disulfide), which serves as a short-term lubricant and an aid to "breaking in" the bearings.

Many bearings are made by attaching some type of plastic lining to a metallic jacket. These linings are produced from a wide variety of plastic materials, the majority being PTFE-based due to PTFE's low-friction and chemical-resistance characteristics. In this type of design, the adhesive that holds the lining in place must be resistant to the service environment in order to prevent detachment of the liner.
A current trend is the increasing use of bearings made exclusively from non-metallics, specifically some of the stronger, temperature-resistant plastics. These are generally superior to metallic bearings from a wear standpoint, and generally also provide lower friction. The materials used vary greatly, but include PEEK, polyimide, and PBI. Whereas these materials have generally good corrosion resistance, their applicability in specific environments must be evaluated before use.

Common Rotary Valve Trim Material Combinations:

The available trim material combinations depend strongly upon the particular style of valve (butterfly, eccentric disk, segmented ball, full ball, plug, etc.). However, there are certain common materials which are offered throughout these styles.

Butterfly disks, eccentric disks, and full balls often start with carbon steel, with either chromium plating or electroless nickel coating. The next step up is a 300-series stainless steel, generally chromium plated or electroless nickel coated. For erosive services, hardsurfacing can be employed on either part or all of the sealing surface. For more erosive applications, solid, cast cobalt-base alloy 6 is often utilized. In plug valves and certain full-ball valves, ceramic trim components are offered for very erosive services. For corrosive services, nickel alloys are utilized. Generally, the wear-resistant coatings and hardfacing materials are not resistant to the environments which dictate the use of nickel alloys, so they must generally be utilized in the uncoated condition along with a non-metallic seal.

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Common shaft, pin, and key materials are strain-hardened type 316, precipitation-hardened type 17-4, and strain-hardened S20910 (commonly referred to as Nitronic® 50). When particular service conditions warrant, these components are also produced from precipitation-hardened or strain-hardened nickel-base alloys.

Seals are made from a very wide variety of materials. Many of the non-metallic seals are made from virgin or filled PTFE, or plastic alloys based upon PTFE. PTFE is so widely used because it has excellent resistance to a broad range of chemicals and a very low friction coefficient. These seals are sometimes solid plastic. Others are metal seals with a captured plastic seat surface.

Many butterfly valves are produced with elastomeric seals, which provide very tight shutoff. The main drawback to elastomeric seals is that a large number of elastomeric materials must be offered to effectively cover a wide variety of chemical environments and temperature ranges.

Metal seals are usually made from 300-series austenitic stainless steels. In many cases, coatings or diffusion treatments (such as nitriding) are utilized to improve wear resistance. Some seals even incorporate a hardsurfaced seat area. Metal seals can be produced in many of the stainless steel and nickel alloys, although their performance may vary due to the differences in galling resistance, wear resistance, and friction coefficients.

Common bearing materials include hardened 440C, solid cobalt-base alloy 6, and type 316 stainless steel with plastic lining. For very corrosive applications, nickel-alloy jackets can be supplied with PTFE based liners and special adhesives. Solid PEEK bearings have recently become popular due to their good wear performance and corrosion resistance.

Valve Packing

This discussion deals with the various materials and systems used by valve manufacturers for valve packing and some of the conditions for selecting a particular packing type. Packing materials have been greatly affected by two events in recent history.

The first event impacted asbestos in the early 1980's. Asbestos had enjoyed the majority of the valve packing market for decades until its classification by the Occupational Safety and Health Administration (OSHA) as a carcinogen and subsequent discouragement for use as an industrial material. Since then, various fibers such as carbon, glass, polyaramid, PTFE, and polybenzimidazole have been used to substitute for asbestos with varying success. Asbestos's unique chemical and heat resistance properties make it irreplaceable in some environments.

The second event was the passage of the Environmental Protection Agency's "Clean Air Act" in 1990. This piece of legislation dramatically reduces the amount of leakage allowed from valve stems by process industries and is creating new requirements for packing materials of construction and configuration. Packing materials vary greatly in content and configuration. The major materials of construction include asbestos, graphite, polytetrafluoroethylene (PTFE), fiberglass, polyaramid, polybenzimidazole and elastomers. These materials can be further broken down into different morphologies such as braided fiber, molded rings, sheets and combinations of any of these.

The bulk of the packing requirements for new control valves are met by two packing types: (1) PTFE V-rings and (2) graphite. The graphite packing can be further subdivided into laminated, ribbon and braided filament packing.

PTFE V-Ring Packing:

This packing is composed of solid rings of molded PTFE (polytetrafluoroethylene). Generally, in a given packing set there are 2 or more packing rings with a "V" cross-section, a male adapter and a female adapter. The packing is generally used over a temperature range of from cryogenic temperatures to 450°F and for all

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chemicals except molten alkali metals and certain fluorine compounds (e.g., fluorine gas or liquid, chlorine trifluoride, oxygen difluoride, etc.). Also, this packing should not be used for nuclear service where the radiation level will exceed 1×10^4 rads.

PTFE is the packing of choice for almost universal chemical compatibility, low maintenance, low leak rates, and minimal cost. This packing can be used with a spring (live loaded) or as jam type packing. The jam type packing requires adjustment of the packing gland during the life of the packing, to make up for wear and relaxation. The relative stem friction is low, but this packing requires relatively smooth stem finish (on the order of the 2 to 4 Ra) . Although rougher stem finishes are utilized successfully in some applications, the additional roughness can provide a means to "pump" process fluid past the packing. This can result in leakage and, in some cases, stem corrosion due to exposure of aerated solution. PTFE packing is the preferred packing for most valve applications.

Graphite/Carbon Packing:

Graphite and / or carbon packing systems are used mainly for valves at temperatures above 450°F (the maximum temperature for PTFE packing). The difference between graphite and carbon is morphology. Carbon is basically amorphous, meaning its atoms are randomly ordered. Graphite is crystalline, which means its atoms are ordered in a precise, repetitive fashion that form crystals. Compared to PTFE, graphite and carbon have a higher initial cost, require more maintenance and produce much more stem friction when loaded sufficiently to meet low leak rate specifications. But for high temperature or fire safe applications, they are the material of choice for sealability and negligible stem wear. Graphite generally has a temperature range from 0° to 1000°F in non-oxidizing service or from 0° to 700°F in oxidizing service. The material can be certified to contain less than 50 ppm of leachable chlorides and halogens and can be used in radioactive nuclear service up to 1.5×10^9 rads total Gamma radiation.

Graphite packing is available in many forms including braided filaments, flexible graphite sheet laminate or ribbon wound die molded rings as well as solid carbon / graphite rings. These materials are typically used in some combination as a composite set. The graphite laminate rings are usually die cut from thin layers of flexible graphite sheet, bonded and cured and then compacted in a die to densify and provide dimensional accuracy. The ribbon wound packing is made similarly except a thin strip is wound onto a mandrel before molding in a die.

Graphite or carbon filament packings are made from a special filament yarn with an interlaced braided construction. Sometimes a PTFE coating is applied during braiding to facilitate construction and provide lubrication in service. At elevated temperatures, above 600°F (316°C), this PTFE coating sublimes, but since its percent volume is relatively small, the packing gland load is not measurably reduced. Filament rings are many times used as end rings in conjunction with the laminated or ribbon wound rings to add compliance to the stack and act as wipers. Carbon and / or graphite braided filament is also used as packing by itself. Braided packing is much more compliant than other solid constructions and is more forgiving as a maintenance packing when the valve stem surface has been damaged mechanically or from corrosion, however it may not seal as well as the solid rings. Hard carbon rings are also used as end rings to act as wipers and anti-extrusion barriers.

Pitting of stainless steel stems has been experienced in the area of contact with graphite packing when valves were wetted during hydrostatic testing or stored in condensing environments. Pitting occurred by a galvanic corrosion mechanism. Carbon and graphite are more "noble" or cathodic on the galvanic series of materials than almost all metals. Laboratory tests have shown that all the stainless steels and even some of the high nickel alloys are susceptible to this type of pitting attack. Metals which are resistant to pitting are N06625, N10276, N06022, titanium and zirconium. To protect other stem materials, a thin sacrificial zinc washer is sometimes used under each graphite laminate ring with the intent of protecting the valve stem from corrosion. The zinc washer does not completely prevent pitting, but it has been shown to help. Also, corrosion inhibitors can be added to the flexible graphite. Older versions used heavy metal based inhibitors

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such as barium molybdate. Newer materials incorporate environmentally safer inorganic, non-metallic, passivating corrosion inhibitors.

To avoid galvanic corrosion, some valve manufacturers hydrostatically test individual components of the valve assembly, allowing them to dry before installing the graphite packing. Since graphite is normally used at elevated service temperatures well above the dew point, moisture is not present to allow galvanic corrosion to start. The continual stroking of the valve in service also reduces the likelihood corrosion.

Gaskets

The materials of construction for gaskets are too numerous to recount, but the main offerings will be described here. The most common flat sheet gaskets include a variety of materials including, elastomer with or without fabric reinforcement, PTFE, asbestos, aramid / rubber, metal and flexible graphite. Spiral wound gaskets are increasingly specified by valve customers. They offer improved "blow-out" protection by virtue of their composite construction. The most common constructions are of PTFE, flexible graphite or an inorganic mineral paper that is wound into spiral laminations with a metal ring encasing the inner and outer diameters. Each material has its own application niche that is some trade-off of performance properties and cost. Key performance properties include temperature resistance, process fluid resistance, sealability, creep relaxation, compressibility, recovery and tensile strength.

Elastomeric Gaskets

Elastomer or synthetic rubber gaskets require very little flange loads to effect a seal. They have very low permeation rates to even small molecule media. They are elastic and can be stretched over a projection during installation without breaking. Elastomer gaskets can also be fabric reinforced to improve the burst strength or blowout resistance. Elastomer gaskets are available in a variety of compositions such as nitrile, neoprene, fluoroelastomer, silicone and ethylene-propylene.

PTFE Gaskets

Polytetrafluoroethylene (PTFE) is most often applied for its excellent chemical resistance. It is a relatively soft plastic that conforms easily to flange surfaces and effects a seal easily. PTFE is more permeable than most gaskets, but its main drawback is its low creep strength. PTFE gaskets tend to creep or cold flow over time and need special care to design and maintain flange loads that don't overload it.

Asbetos Gaskets

Asbestos was the general gasket material of choice for decades until its fibers were linked to a respiratory ailment named asbestosis. The Occupational Safety and Health Act legislated work rules that encouraged its disuse in the early 1980's. Since then, use of asbestos gasketing has been almost nil. There are some environments that still warrant its use, such as high temperature oxidizing agents. Asbestos is excellent for almost all steam applications, hydrocarbons and a vast array of chemicals. It can handle high flange loads without creep relaxation and is inexpensive to produce. Gasket asbestos is usually composed of metal silicate minerals called crocidolite or chrysotile or a combination of both with an elastomeric binder compatible with the process fluid. Asbestos remains a popular gasket material in some countries, but not in the United States.

Aramid Gaskets

Aramid fiber gaskets with various elastomer binders became the asbestos replacement material of choice in the 1980's. Aramids are special high temperature aromatic polyamides that have exceptional strength to weight ratios and unusually high (stiff) flexural moduli. The reinforcing aramid fibers, like asbestos, are bound together in a flexible sheet gasket with an assortment of elastomers that must be specified to resist the process fluid. These gaskets generally seal better than the asbestos gaskets they replace, but have less chemical and temperature resistance.

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Metal Gaskets

Metal gaskets have absolute sealing capabilities, requiring high bolt loads, but need extremely flat, smooth flange surfaces to effect a seal. Dead soft, annealed sheet materials are used as gaskets. Usual alloys are UNS S31600 and N04400. Silver plating on both sides of the gasket is also specified to improve sealability and avoid crevice corrosion.

Flexible Graphite Gaskets

Flexible graphite is a unique material that dominates the high temperature gasket market today. Composed of all carbon graphite and very few contaminants, flexible graphite has excellent chemical resistance to all but the strongest oxidizers. It has excellent compliance and sealing capabilities with low to high bolt loads. It can be purchased as a laminate with thin stainless steel shims to improve its handling ruggedness before assembly, or as pure graphite to eliminate chemical compatibility concerns. It also is available in grades with extra low halogen, sulfur, nitrates and low melting metals to comply with nuclear power industry specifications. Graphite also has almost universal chemical resistance with the exception of strong oxidizing compounds.

O-ring Seals

An O-ring is a toroid-shaped object usually made from an elastomer which is mechanically compressed inside a gland or tightly dimensioned groove which effects a seal for a circular shaped leak path. O-rings are also made from plastics or metals for special applications; however, the elastic recovery of elastomers after deformation make them especially suited to this application. They can also be made to have cross sections that are not circular for special applications. For instance, square or "T" cross section rings are sometimes used as dynamic seals at elevated temperatures so that they are less apt to roll and suffer "spiraling" failures.

O-rings provide a means for low cost, extremely leak tight seals. O-rings are generally used as diametral (i.e., the gap between a mating piston and cylinder bore) or face (i.e., the gap between two flat, parallel surfaces) seals. They can be used as static or dynamic seals with varying requirements for compressive preload depending upon the pressure and composition of the medium to be sealed. Because of their symmetry, O-rings can be used as bi-directional seals, i.e., the pressure differential can alternate from one side of the O-ring to the other.

The materials O-rings are molded from must be carefully selected to be compatible with the medium to be sealed and with the temperature of service. Sometimes, slight swelling of the O-ring in its application medium can be used to improve sealing characteristics without designing glands with large compressive preloads. Other considerations include the exposure medium on the non-pressurized side of the O-ring. For instance, some oil-resisting elastomers such as nitrile don't have good weathering resistance in air. Also, the rate at which pressure is reduced in a high pressure application can cause explosive decompression in the O-ring. This is caused when gas or fluid medium that has permeated the O-ring material under high pressure conditions rapidly exits the material when pressure is reduced and causes mechanical tearing of the O-ring.

Spring Energized, Pressure Assisted Seals

Spring energized, pressure assisted seals (SEPAS's) are sealing devices consisting of a PTFE or other polymeric jacket partially covering a corrosion resistant metal spring energizer. The polymeric jacket is designed to be compliant, non-galling, low friction and resistant to a wide array of chemicals. The spring energizers have a variety of configurations to supply different amounts of load to the polymeric jacket lips and effect a seal against the gland walls. Alloy compositions also vary depending upon the fluid medium and temperature. O-rings can also be used as seal energizers. SEPAS's have a degree of unidirectionality. Since they are pressure assisted, the orientation of the SEPAS is critical to allow pressure assist to improve sealing across a wide range of pressures. SEPAS can be used as diametral (i.e., the gap between a mating piston and cylinder bore) or face (i.e., the gap between two flat, parallel surfaces) seals. They can be used as static or dynamic seals with varying specifications for energizer loads depending upon requirements for friction, seal integrity and seal life.

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While higher in cost than an O-ring, spring energized, pressure assisted seals provide excellent seal integrity with wider temperature ranges, more universal chemical resistance, and fewer shelf life issues than O-rings.

Sealants

Sealants are used in a number of surface interfaces or joints in valve assemblies, but predominantly to seal threads and especially pipe threads. A vast number of product compositions and brand names are available, but they can be quickly reduced to purely polymeric and metal dispersion types.

Polymeric Adhesives

The polymeric adhesives include a variety of materials, perhaps best separated by their cure systems, chemically reactive (thermosetting), evaporation or diffusion, and hot melt (thermoplastic). The chemically reactive sealants usually require mixing of two parts before using, but may be premixed and only require moisture from air or heat to cure. These materials include epoxies, polyurethanes, phenolics, polyimides, silicones, cyanoacrylates, modified acrylics, and phenolics. These materials have a wide variety of properties and are good choices for applications which are limited to temperatures between 100°C and 260°C (212 and 500°F). The higher temperature capabilities are accomplished with polyimide, phenolics and epoxies. Anaerobic sealants are a special classification of chemically reactive polymers in that they are liquid monomers that cure by free radical polymerization in the absence of oxygen. These sealants are very common thread sealants as they cure when oxygen is eliminated in the joint during tightening. They are also commonly loaded with a polytetrafluoroethylene (PTFE) dispersion to improve their heat resistance (at or above 260°C, 500°F) and general chemical resistance. These materials satisfy the requirements of most instrument applications that require sealants on pneumatic tubes and fittings or assembled components such as bellows, diaphragms and nozzles. Also, low temperature valve assembly applications such as body drain plugs, seat ring retainers, etc. are appropriately sealed with these materials.

Evaporation or diffusion cured systems include natural and synthetic rubber, polyurethanes, vinyls, acrylics, and some phenolics. They require no mixing but may require heat to drive off solvent or water-based diluents. These materials see similar applications as the chemically reactive materials listed above.

The hot melt or thermoplastic sealants or adhesives are generally used only during fabrication of subassemblies in instruments or perhaps lined valve bearings. They are generally limited in application by their melting point which is generally 100 to 150°C (212 to 302°F) for polyamide or polyester materials.

Metallic Dispersions

The metallic dispersions are necessary for high temperature applications and are usually in a polymeric based sealant or dispersed in a thick hydrocarbon which allows sufficient tack to adhere to components during assembly. The metal flake or particulate content is such a high percentage that even when elevated temperatures bake out the hydrocarbon or polymer, the soft metal particles still effect a seal with little volume lost. Metal flake or particles commonly used are lead, zinc, copper and nickel. However, there are some concerns in applying these materials to high strength steels and stainless steels at elevated temperatures.

Lead and zinc in particular have been documented to cause liquid metal and even solid metal embrittlement of hardened steels. Lead can cause liquation cracking of stainless steels or hardened alloy steels at temperatures as low as 260°C (500°F). Zinc can cause solid metal embrittlement of stainless and alloy steels in the quenched and tempered condition when exposed to temperatures as low as 260°C (500°F). Zinc can also cause liquation cracking of austenitic stainless steels at temperatures above its melting point at 420°C (787°F). There are no documented cases of copper or nickel embrittlement of steels known to this author.

Fortunately, code approved pressure retaining materials such as valve bodies are generally not in a quenched and tempered condition, so the use of zinc sealants on drain plugs or threaded seat rings is not a problem as long as application temperatures remain below the melting point of zinc (420°C or 787°F). Most valves are

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applied below this temperature. However, studs and bolting are usually in a high strength heat treated condition, and contact with zinc or lead containing sealants could be a problem at temperatures as low as 260°C (500°F). When in doubt of a valve's application temperature, it is best to use a sealant that utilizes copper or nickel metal.

Paint and External Coatings

Valves see a variety of service conditions as installed which vary from well controlled indoor environments to extremely aggressive industrial environments where chemical exposure is continuous. For this reason, a variety of external coatings are employed to protect the exterior of valves and accessories. Protective coatings for valve exteriors are usually referred to as "paint" by laymen, but those involved with the technology usually refer to them as coatings.

Pretreatment

Most metals are pretreated before coating. Pre-treating includes some form of cleaning and conversion coating before coating. Metals may simply be solvent or detergent cleaned, but if steel castings have been stored outside either by the manufacturer or the foundry, they will need to be abrasively grit blasted before dip or spray cleaning. Also, ferrous materials are usually phosphate conversion coated followed by a chromate conversion coating to impart corrosion protection and provide a surface with "tooth" for good adhesion of the final coating. Aluminum based materials are usually chromate conversion coated for the same purpose.

The following is a synopsis of the usual coatings available from original equipment valve manufacturers and/or preferred by valve customers. Other coatings are available as options from manufacturers, but special customer order coatings are difficult to comply with as high volume coating equipment is specially suited to a particular standard coating system and require retrofit to apply other coating materials. Also, it is difficult to acquire specialized coatings and associated Material Safety Data Sheets (MSDS) and get the appropriate plant safety and environmental management personnel approvals and personnel protection equipment in place before the valves need be delivered. This holds especially true for solvent based paints.

Alkyds

For light industrial environments, most valve equipment is supplied with an alkyd (synthetic resin) coating either with or without a primer. Alkyds have good gloss and color retention, good wetting and penetration characteristics as well as good outdoor weathering characteristics. They are most often spray applied, but have very limited solvent, water and alkali resistance.

Acrylic Latex

For moderate industrial environments, the usual coating for valve, actuator and mounted accessories, e.g. positioner and air set regulator, is an acrylic latex. Acrylic latexes can be spray applied, have good adhesion and moderate chemical resistance. The coating is also nonflammable and meets volatile organic compound (VOC) regulations. Acrylic latex is compatible with zinc rich primers to improve corrosion resisting properties.

Primers containing zinc offer not only barrier resistance like any other coating, but also offer anodic protection when the coating is perforated. The zinc is more anodic than steels on the galvanic series and therefore corrodes preferentially to the steel base metal. Thus, zinc provides sacrificial corrosion protection.

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Epoxies and Polyesters

For more aggressive environments where continuous chemical vapor and occasional chemical wetting of the valve equipment occurs, polyester or epoxy resin coatings are usually applied. There are a variety of epoxies available on the market. The two most commonly associated with valves can be either catalyzed with amines or polyamides or they are coal tar epoxies. Epoxies are usually two component, high-build coatings that can be liquid spray applied or dry powder coated and heat cured. Epoxies have excellent solvent, water and alkali resistance. However, they tend to be brittle when impacted and also chalk when exposed to ultraviolet (UV) light.

Dry powder coated and heat cured polyesters give up very little solvent resistance when compared to epoxies and have the advantages of better performance in salt spray tests and much better UV light resistance. Polyesters are also more flexible and much less apt to chip or spall off when subjected to impacts. Powder coating is capable of extremely high quality coatings. It offers a high-build (thick), dense and continuous coating. Powder coating has the additional processing advantage of no VOC's released into the environment and no paint sludge from overspray to dispose of.

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Inconel® is a trademark of Inco Alloys International, Inc.
Monel® is a trademark of Inco Alloys International, Inc.
Nitronic® is a trademark of Armco Steels Corporation
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TABLE 1 - GENERAL CORROSION DATA **(Courtesy Fisher Controls)**
This corrosion table is only intended to give a general indication of how various materials will react when in contact with certain fluids at ambient temperature. The data cannot be absolute because concentration, temperature, pressure and other conditions may alter the suitability of a particular material. There are also economic considerations that may influence material selection. Use this table as a guide only.

A - Minimal corrosion
B - Minor to moderate effect, proceed with caution
C - Unsatisfactory

Al ..... Aluminum
Br. ..... Brass
Steel ..... Carbon steel, WCB, WCC, LCB, LCC, WC9 and C5
CI ..... Cast iron
416 & 440C..... Also includes 410, CA15 and CA6NM
17-4 ..... Includes 17-4 PH®, CB7Cu-1 and CB7Cu-2
304 ..... Includes 304L, CF3 and CF8
316 ..... Includes 316L, CF3M, CF8M, 317 and CG8M
Duplex ..... Includes 2205, CD3MN, Ferralium® 255, CD7MCuN, CD4MCu and others
254 SMO..... Includes S31254 (Avesta® 254 SMO) and CK3MCuN
20 ..... Includes Carpenter 20Cb-3® and CN7M
400 ..... Includes Monel® 400, R405, M35-1, K500
C276..... Includes Hastelloy® C276, CW2M, C22 and C4
B2..... Includes Hastelloy® B2 and N7M
6 ..... Cobalt-base Stellite® Alloy 6 and CoCr-A
Ti..... Titanium
Zr Zirconium

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AMBIENT TEMPERATURE CORROSION INFORMATION

FLUID	Al.	Br.	CI & Steel	416 & 440C	17-4 SST	304 SST	316 SST	Duplex SST	254 SMO	Alloy 20	Alloy 400	Alloy C276	Alloy B2	Alloy 6	Ti.	Zr.
Acetaldehyde	A	A	C	A	A	A	A	A	A	A	A	A	A	A	A	A
Acetic Acid, Air Free	C	C	C	C	C	C	A	A	A	A	A	A	A	A	A	A
Acetic Acid, Aerated	C	C	C	C	B	B	A	A	A	A	C	A	A	A	A	A
Acetone	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Acetylene	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Alcohols	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Aluminum Sulfate	C	C	C	C	B	A	A	A	A	A	B	A	A	A	A	A
Ammonia	A	C	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Ammonium Chloride	C	C	C	C	C	C	B	A	A	A	B	A	A	B	A	A
Ammonium Hydroxide	A	C	A	A	A	A	A	A	A	A	C	A	A	A	A	B
Ammonium Nitrate	B	C	B	B	A	A	A	A	A	A	C	A	A	A	C	A
Ammonium Phosphate (Mono-Basic)	B	B	C	B	B	A	A	A	A	A	B	A	A	A	A	A
Ammonium Sulfate	C	C	C	C	B	B	A	A	A	A	A	A	A	A	A	A
Ammonium Sulfite	C	C	C	C	A	A	A	A	A	A	C	A	A	A	A	A
Aniline	C	C	C	C	A	A	A	A	A	A	B	A	A	A	A	A
Asphalt	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Beer	A	A	B	B	A	A	A	A	A	A	A	A	A	A	A	A
Benzene (Benzol)	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Benzoic Acid	A	A	C	C	A	A	A	A	A	A	A	A	A	A	A	A
Boric Acid	C	B	C	C	A	A	A	A	A	A	B	A	A	A	A	A
Bromine, Dry	C	C	C	C	B	B	B	A	A	A	A	A	A	A	C	C
Bromine, Wet	C	C	C	C	C	C	C	C	C	C	A	A	A	C	C	C
Butane	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Calcium Chloride	C	C	B	C	C	B	B	A	A	A	A	A	A	A	A	A
Calcium Hypochlorite	C	C	C	C	C	C	C	A	A	A	C	A	B	B	A	A
Carbon Dioxide, Dry	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Carbon Dioxide, Wet	A	B	C	C	A	A	A	A	A	A	A	A	A	A	A	A
Carbon Disulfide	C	C	A	B	B	A	A	A	A	A	B	A	A	A	A	A
Carbonic Acid	A	B	C	C	A	A	A	A	A	A	A	A	A	A	A	A
Carbon Tetrachloride	A	A	B	B	A	A	A	A	A	A	A	A	A	A	A	A
Caustic Potash (see Potassium Hydroxide)																
Caustic Soda (see Sodium Hydroxide)																
Chlorine, Dry	C	C	A	C	B	B	B	A	A	A	A	A	A	A	C	A
Chlorine, Wet	C	C	C	C	C	C	C	C	C	C	B	B	B	C	A	A
Chromic Acid	C	C	C	C	C	C	C	B	A	C	C	A	B	C	A	A
Citric Acid	B	C	C	C	B	B	A	A	A	A	A	A	A	A	A	A
Coke Oven Acid	C	B	A	A	A	A	A	A	A	A	B	A	A	A	A	A
Copper Sulfate	C	C	C	C	C	C	B	A	A	A	C	A	A	C	A	A
Cottonseed Oil	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Creosote	C	C	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Dowtherm	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Ethane	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Ether	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A
Ethyl Chloride	C	B	C	C	B	B	B	A	A	A	A	A	A	A	A	A
Ethylene	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A

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FLUID	Al.	Br.	CI & Steel	416 & 440C	17-4 SST	304 SST	316 SST	Duplex SST	254 SMO	Alloy 20	Alloy 400	Alloy C276	Alloy B2	Alloy 6	Ti.	Zr.
Ethylene Glycol	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Ferric Chloride	C	C	C	C	C	C	C	C	B	C	C	A	C	C	A	A
Fluorine, Dry	B	B	A	C	B	B	B	A	A	A	A	A	A	A	C	C
Fluorine, Wet	C	C	C	C	C	C	C	C	C	C	B	B	B	C	C	C
Formaldehyde	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A
Formic Acid	B	C	C	C	C	C	B	A	A	A	C	A	B	B	C	A
Freon, Wet	C	C	B	C	B	B	A	A	A	A	A	A	A	A	A	A
Freon, Dry	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A
Furfural	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A
Gasoline, Refined	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Glucose	A	A	A	A	A	A	A	C	A	A	A	A	A	A	A	A
Hydrochloric Acid (Aerated)	C	C	C	C	C	C	C	C	C	C	C	B	A	C	C	A
Hydrochloric Acid (Air Free)	C	C	C	C	C	C	C	C	C	C	C	B	A	C	C	A
Hydrofluoric Acid (Aerated)	C	C	C	C	C	C	C	C	C	C	B	B	B	C	C	C
Hydrofluoric Acid (Air Free)	C	C	C	C	C	C	C	C	C	C	A	B	B	C	C	C
Hydrogen	A	A	A	C	B	A	A	A	A	A	A	A	A	A	C	A
Hydrogen Peroxide	A	C	C	C	B	A	A	A	A	A	C	A	C	A	A	A
Hydrogen Sulfide	C	C	C	C	C	A	A	A	A	A	A	A	A	A	A	A
Iodine	C	C	C	C	A	A	A	A	A	A	C	A	A	A	C	B
Magnesium Hydroxide	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Mercury	C	C	A	A	A	A	A	A	A	A	B	A	A	A	C	A
Methanol	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Methyl Ethyl Ketone	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Milk	A	A	C	A	A	A	A	A	A	A	A	A	A	A	A	A
Natural Gas	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Nitric Acid	C	C	C	C	A	A	A	A	A	A	C	B	C	C	A	A
Oleic Acid	C	C	C	B	B	B	A	A	A	A	A	A	A	A	A	A
Oxalic Acid	C	C	C	C	B	B	B	A	A	A	B	A	A	B	C	A
Oxygen	C	A	C	C	B	B	B	B	B	B	A	B	B	B	C	C
Petroleum Oils, Refined	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Phosphoric Acid (Aerated)	C	C	C	C	B	A	A	A	A	A	C	A	A	A	C	A
Phosphoric Acid (Air Free)	C	C	C	C	B	B	B	A	A	A	B	A	A	B	C	A
Picric Acid	C	C	C	C	B	B	A	A	A	A	C	A	A	A	A	A
Potash (see Potassium Carbonate)																
Potassium Carbonate	C	C	B	B	A	A	A	A	A	A	A	A	A	A	A	A
Potassium Chloride	C	C	B	C	C	B	B	A	A	A	A	A	A	A	A	A
Potassium Hydroxide	C	C	B	B	A	A	A	A	A	A	A	A	A	A	A	A
Propane	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Rosin	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A
Silver Nitrate	C	C	C	C	B	A	A	A	A	A	C	A	A	A	A	A
Soda Ash (see Sodium Carbonate)																
Sodium Acetate	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Sodium Carbonate	C	C	A	B	A	A	A	A	A	A	A	A	A	A	A	A
Sodium Chloride	C	A	C	C	B	B	B	A	A	A	A	A	A	A	A	A
Sodium Chromate	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Sodium Hydroxide	C	C	A	B	B	B	A	A	A	A	A	A	A	A	A	A
Sodium Hypochlorite	C	C	C	C	C	C	C	C	C	C	C	A	B	C	A	A
Sodium Thiosulfate	C	C	C	C	B	B	A	A	A	A	A	A	A	A	A	A
Stannous Chloride	C	C	C	C	C	C	B	A	A	A	C	A	A	B	A	A
Steam	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A

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FLUID	Al.	Br.	CI & Steel	416 & 440C	17-4 SST	304 SST	316 SST	Duplex SST	254 SMO	Alloy 20	Alloy 400	Alloy C276	Alloy B2	Alloy 6	Ti.	Zr.
Stearic Acid	C	B	B	B	B	A	A	A	A	A	A	A	A	B	A	A
Sulfate Liquor (Black)	C	C	A	C	C	B	A	A	A	A	A	A	A	A	A	A
Sulfur	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Sulfur Dioxide, Dry	C	C	C	C	C	C	B	A	A	A	C	A	A	B	A	A
Sulfur Trioxide, Dry	C	C	C	C	C	C	B	A	A	A	B	A	A	B	A	A
Sulfuric Acid (Aerated)	C	C	C	C	C	C	C	A	A	A	C	A	C	B	C	A
Sulfuric Acid (Air Free)	C	C	C	C	C	C	C	A	A	A	B	A	A	B	C	A
Sulfurous Acid	C	C	C	C	C	B	B	A	A	A	C	A	A	B	A	A
Tar	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Trichloroethylene	B	B	B	B	B	B	A	A	A	A	A	A	A	A	A	A
Turpentine	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A
Vinegar	B	B	C	C	A	A	A	A	A	A	A	A	A	A	A	A
Water, Boiler feed, Amine Treated	A	A	A	A	A	A	A	A	A	A	A	A	A	C	A	A
Water, Distilled	A	A	C	C	A	A	A	A	A	A	A	A	A	A	A	A
Water, Sea	C	A	C	C	C	C	B	A	A	A	A	A	A	A	A	A
Whiskey and Wines	A	A	C	C	A	A	A	A	A	A	A	A	A	A	A	A
Zinc Chloride	C	C	C	C	C	C	C	B	B	B	A	A	A	B	A	A
Zinc Sulfate	C	C	C	C	A	A	A	A	A	A	A	A	A	A	A	A

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Table 2 - FLUID COMPATIBILITY TABLE

Fluid	Elastomer Ratings For Compatibility With Fluid KEY: + = Best Possible Selection A = Generally Compatible B = Marginally Compatible C = Not Recommended **NOTE:** These recommendations are to be used as a *general guide only.* Full details regarding pressure, temperature, chemical considerations, and the mode of operation must be ed when selecting an elastomer.											
	ACM, ANM Polyacrylic	AU, EU Polyurethane	CO, ECO Epichloro-hydrin	CR Chloroprene Neoprene	EPM, EPDM Ethylene Propylene	FKM Fluoro-elastomer Viton	FFKM Perfluoro-elastomer	IIR Butyl	MQ, PMQ, VMQ, PVMQ Silicone	NBR Nitrile Buna N	NR Natural Rubber	TFE/P Tetrafluoroethylene-propylene copolymer
Acetic Acid (30%)	C	C	C	C	A+	C	A+	A	A	B	B	C
Acetone	C	C	C	C	A	C	A	A	C	C	C	C
Air, ambient	A	A	-	A	A	A	A	A	A	A	B	A
Air, hot (200°F)	B	B	-	C	A	A	A	A	A	A	B	A
Air, hot (400°F)	C	C	-	C	C	A	A	C	A	C	C	A
Alcohol, ethyl	C	C	-	A	A	C	A	A	A	A	A	A
Alcohol, methyl	C	C	B	A+	A	C	A	A	A	A	A	A
Ammonia, anhydrous, liquid	C	C	-	A+	A	C	A	A	B	B	C	A
Ammonia, gas (hot)	C	C	-	B	B	C	A	B	A	C	C	A+
Beer (beverage)	C	C	A	A	A	A	A	A	A	A	A	A
Benzene	C	C	C	C	C	A	A	C	C	C	C	C
Black Liquor	C	C	-	B	B	A+	A	C	C	B	B	A
Blast Furnace Gas	C	C	-	C	C	A+	A	C	A	C	C	A
Brine (calcium chloride)	A	A	A	A	A	A	A	A	A	A	A	A
Butadiene Gas	C	C	C	C	C	A+	A	C	C	C	C	-
Butane Gas	A	C	A	A	C	A	A	C	C	A+	C	B
Butane, liquid	A	C	A	B	C	A	A	C	C	A	C	C
Carbon Tetrachloride	C	C	B	C	C	A+	A	C	C	C	C	C
Chlorine, dry	C	C	B	C	C	A+	A	C	C	C	C	C
Chlorine, wet	C	C	B	C	C	A+	A	C	C	C	C	B
Coke Oven Gas	C	C	-	C	C	A+	A	C	B	C	C	A
Dowtherm A	C	C	C	C	C	A+	A	C	C	C	C	B
Ethyl Acetate	C	C	C	C	B	C	A	B	B	C	C	C
Ethylene Glycol	C	B	A	A	A+	A	A	A	A	A	A	A
Freon 11	A	C	-	C	C	B+	B	C	C	B	C	C
Freon 12	B	A	A	A+	B	B	B	B	C	A	B	C
Freon 22	B	C	A	A+	A	C	A	A	C	C	A	C
Freon 114	-	A	A	A	A	A	B	A	C	A	A	C

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Elastomer Ratings For Compatibility With Fluid

KEY: + = Best Possible Selection A = Generally Compatible B = Marginally Compatible C = Not Recommended

NOTE: These recommendations are to be used as a *general guide only.* Full details regarding pressure, temperature, chemical considerations, and the mode of operation must be ed when selecting an elastomer.

Fluid	ACM, ANM Polyacrylic	AU, EU Polyurethane	CO, ECO Epichloro-hydrin	CR Chloroprene Neoprene	EPM, EPDM Ethylene Propylene	FKM Fluoro-elastomer Viton	FFKM Perfluoro-elastomer	IIR Butyl	MQ, PMQ, VMQ, PVMQ Silicone	NBR Nitrile Buna N	NR Natural Rubber	TFE/P Tetrafluoroethylene-propylene copolymer
Freon Replacements (see Suva)												
Gasoline	C	B	A	C	C	A	A	C	C	A+	C	C
Hydrogen Gas	B	A	-	A	A	A	A	A	C	A	B	A
Hydrogen Sulfide (dry)	C	B	B	A	A+	C	A	A	C	A	A	A
Hydrogen Sulfide (wet)	C	C	B	A	A+	C	A	A	C	C	C	A
Jet Fuel (JP-4)	B	B	A	C	C	A	A	C	C	A	C	B
Methylene Chloride	C	C	-	C	C	B+	A+	C	C	C	C	B
Milk	C	C	-	A	A	A	A	A	A	A+	A	A
Naphthalene	-	B	-	C	C	A+	A	C	C	C	C	B
Natural Gas	B	B	A	A	C	A	A	C	C	A+	B	A
Natural Gas+H_2S (Sour Gas)	C	B	A	A+	C	C	A	C	C	B	C	A
Natural Gas, Sour + Ammonia	C	C	-	B+	C	C	A	C	C	B	C	A+
Nitric Acid (10%)	C	C	C	C	B	A+	A	A	C	C	C	A
Nitric Acid (50-100%)	C	C	C	C	C	A+	A	A	C	C	C	B
Nitric Acid Vapor	C	C	C	B	B	A	A	B	C	C	C	A
Nitrogen	A	A	A	A	A	A	A	A	A	A	A	A
Oil (fuel)	B	C	A	B	C	A	A	C	C	A+	C	A
Ozone	B	A	A	B	A	A	A	B	A	C	C	A
Paper Stock	-	C	-	B	B	A	A	B	C	B	C	-
Propane	A	B	A	A	C	A	A	C	C	A+	C	A
Sea Water	C	B	-	B	A	A	A	A	A	A	B	A
Sea Water + Sulfuric Acid	C	B	-	B	B	A	A	B	C	C	C	A
Soap Solutions	C	C	A	A	A	A	A	A	A	A	B	A
Steam	C	C	C	C	B+	C	A	B	C	C	C	A+
Sulfur Dioxide (dry)	C	-	-	C	A+	-	-	B	B	C	B	-
Sulfur Dioxide (wet)	C	B	-	B	A+	C	A	A	B	C	C	B
Sulfuric Acid (to 50%)	B	C	B	C	B	A+	A	C	C	C	C	A
Sulfuric Acid (50-100%)	C	C	C	C	C	A+	A	C	C	C	C	A

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Fluid	**Elastomer Ratings For Compatibility With Fluid** KEY: + = Best Possible Selection A = Generally Compatible B = Marginally Compatible C = Not Recommended **NOTE:** These recommendations are to be used as a *general guide only.* Full details regarding pressure, temperature, chemical considerations, and the mode of operation must be ed when selecting an elastomer.											
	ACM, ANM Polyacrylic	AU, EU Polyurethane	CO, ECO Epichloro-hydrin	CR Chloroprene Neoprene	EPM, EPDM Ethylene Propylene	FKM Fluoro-elastomer Viton	FFKM Perfluoro-elastomer	IIR Butyl	MQ, PMQ, VMQ, PVMQ Silicone	NBR Nitrile Buna N	NR Natural Rubber	TFE/P Tetrafluoroethylene-propylene copolymer
Suva HCFC-123	-	C	-	A+	A+	B	-	A+	B	C	C	-
Suva HFC-134a	-	-	-	B	A	C	-	B	B	A+	B	-
Water (ambient)	C	C	B	A	A	A	A	A	A	A	A	A
Water (200°F)	C	C	B	C	A+	B	A	B	A	C	A	-
Water (300°F)	C	C	-	C	B+	C	A	B	C	C	C	-
Water (de-ionized)	C	A	-	A	A	A	A	A	A	A	A	A
Water, white	C	B	-	B	A	A	A	A	B	B	B	-

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Table 3 - GENERAL PROPERTIES OF ELASTOMERS

Property	ACM, ANM (1)Polyacrylate	AU, EU (2)Polyurethane	CO, ECO Epichlorohydrin	CR Chloroprene	EPM, EPDM (3)Ethylene Propylene	FKM (1, 2) Fluorocarbon	FFKM Perfluorocarbon	IIR Butyl	MQ, PMQ, VMQ, PVMQ Silicone	NBR Nitrile Buna-N	NR Natural Rubber	SBR Buna-S GRS	TFE/P Tetrafluoroethylene/Propylene
Tensile: Pure Gum	100	....	2000	3500	....	....	....	3000	200-450	600	3000	400	....
Reinforced	1800	6500	2500	3500	2500	2300	3200	3000	1100	4000	4500	3000	2800
Tear Resistance	Fair	Excellent	Good	Good	Poor	Good	....	Good	Poor-Fair	Fair	Excellent	Poor-Fair	Good
Abrasion Resistance	Good	Excellent	Fair	Excellent	Good	Very Good	....	Fair	Poor	Good	Excellent	Good	Good
Aging: Sunlight	Excellent	Excellent	Good	Excellent	Excellent	Excellent	Excellent	Excellent	Good	Poor	Poor	Poor	....
Oxidation	Excellent	Excellent	Good	Good	Good	Excellent	Excellent	Good	Very Good	Fair	Good	Fair	Excellent
Heat: (Max. Temp.)	350°F	200°F	275°F	200°F	350°F	400°F	550°F	200°F	450°F	250°F	200°F	200°F	400°F
Flex Cracking Resistance	Good	Excellent	....	Excellent	....	....	....	Excellent	Fair	Good	Excellent	Good	....
Compression Set Resistance	Good	Good	Fair	Excellent	Fair	Poor	....	Fair	Good	Very Good	Good	Good	Good
Solvent Resistance:													
Aliphatic Hydrocarbon	Good	Very Good	Excellent	Fair	Poor	Excellent	Excellent	Poor	Poor	Good	Very Poor	Very Poor	Good
Aromatic Hydrocarbon	Poor	Fair	Good	Poor	Fair	Very Good	Excellent	Very Poor	Very Poor	Fair	Very Poor	Very Poor	Fair
Oxygenated Solvent	Poor	Poor	....	Fair	....	Good	Excellent	Good	Poor	Poor	Good	Good	Poor
Halogenated Solvent	Poor	....	....	Very Poor	Poor	....	Excellent	Poor	Very Poor	Very Poor	Very Poor	Very Poor	Poor/Good
Oil Resistance:													
Low Aniline Mineral Oil	Excellent	....	....	Fair	Poor	Excellent	Excellent	Very Poor	Poor	Excellent	Very Poor	Very Poor	Excellent
High Aniline Mineral Oil	Excellent	....	....	Good	Poor	Excellent	Excellent	Very Poor	Good	Excellent	Very Poor	Very Poor	Fair
Synthetic Lubricants	Fair	....	Excellent	Very Poor	Poor	....	Excellent	Poor	Fair	Fair	Very Poor	Very Poor	Excellent
Organic Phosphates	Poor	Poor	Excellent	Very Poor	Very Good	Poor	Excellent	Good	Poor	Very Poor	Very Poor	Very Poor	Good
Gasoline Resistance:													
Aromatic	Fair	Fair	Excellent	Poor	Fair	Good	Excellent	Very Poor	Poor	Good	Very Poor	Very Poor	Poor
Non-Aromatic	Poor	Good	Excellent	Good	Poor	Very Good	Excellent	Very Poor	Good	Excellent	Very Poor	Very Poor	Fair
Acid Resistance:													
Dilute (Under 10%)	Poor	Fair	Good	Fair	Very Good	Excellent	Excellent	Good	Fair	Good	Good	Good	Excellent
^Concentrated	Poor	Poor	Good	Fair	Good	Very Good	Excellent	Fair	Poor	Poor	Fair	Poor	Good
Low Temperature Flexibility (Max.)	-10°F	-40°F	-40°F	-40°F	-50°F	-30°F	0°F	-40°F	-100°F	-40°F	-65°F	-50°F	0°F
Permeability to Gases	Good	Good	Excellent	Very Good	Good	Good	Fair	Very Good	Fair	Fair	Fair	Fair	....
Water Resistance	Fair	Fair	Fair	Fair	Very Good	Excellent	Excellent	Very Good	Fair	Very Good	Good	Very Good	Excellent
Alkali Resistance:													
Dilute (Under 10%)	Poor	Fair	Excellent	Good	Excellent	Excellent	Excellent	Very Good	Fair	Good	Good	Good	Excellent
Concentrated	Poor	Poor	Excellent	Good	Good	Very Good	Excellent	Very Good	Poor	Fair	Fair	Fair	Good
Resilience	Very Poor	Fair	Fair	Very Good	Very Good	Good	....	Very Good	Good	Fair	Very Good	Fair	....
Elongation (Max.)	200%	625%	400%	500%	500%	425%	142%	700%	300%	500%	700%	500%	400%

^Except Nitric and Sulfuric.
NOTES: 1. Do not use with steam. 2. Do not use with ammonia. 3. Do not use with petroleum base fluids. Use with ester Base (non-Flammable) hydraulic oils and low pressure steam applications to 300°F.

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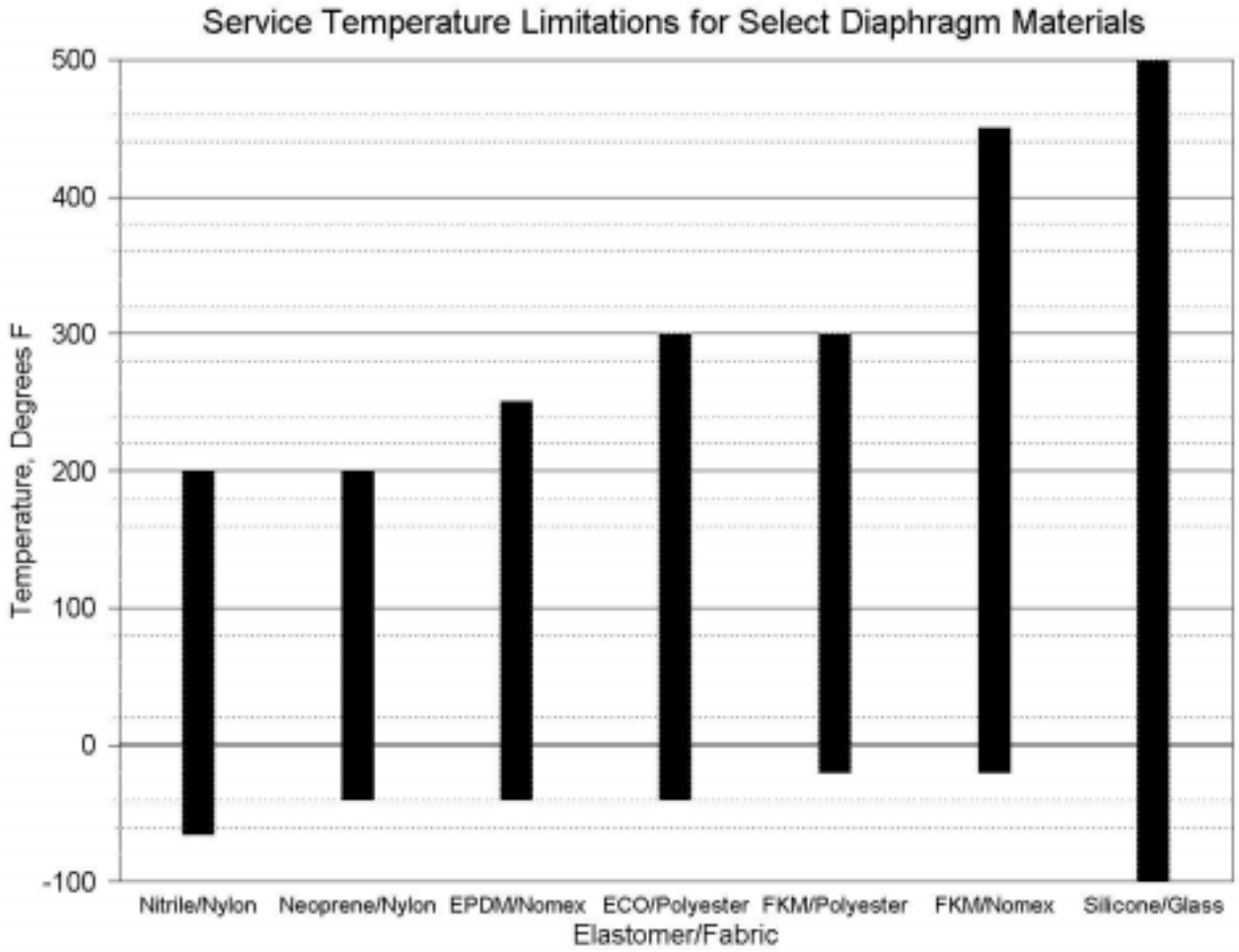


Figure 1 Temperature Ratings for Common Diaphragm Materials
(Courtesy Fairprene Industrial Product Company)

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Table 4 - Common Pressure Retaining Materials in Various Forms

Nominal Composition	Forgings	Castings	Plate
C-Mn-Si, low temperature, -50°F (-46°C)	A350 grades LF1 and LF2	A352 LCB and LCC	A537 with impact testing
C-Mn-Si	A105	A216 WCC and WCB	A515 grade 70 and A516 grade 70
2¼Cr-1Mo	A182 grade F22	A217 WC9	A387 grade 22 class 1 and 2
5Cr-½Mo	A182 grades F5 and F5a (501 SST)	A217 C5	A387 grade 5 class 1
304L	A182 F304L (S30403)	A351 CF3	A240 S30403
316	A182 F316 (S31600)	A351 CF8M	A240 S31600
317	A182 F317 (S31700)	A351 CG8M	A240 S31700
347	A182 F347 (S34700)	A351 CF8C	A240 S34700
254 SMO®	A182 F44 (S31254)	A351 CK3MCuN	A240 S31254
Carpenter 20Cb-3®	B462 N08020	A351 CN7M	B463 N08020
Nickel 200	B564 N02200	A494 CZ100	B162 N02200
Monel® 400	B564 N04400	A494 M35-1	B127 N04400
Inconel® 600	B564 N06600	A494 CY40	B168 N06600
Hastelloy® B2	B335 N10665	A494 N7M	B333 N10665
Hastelloy® C	B574 N10276	A494 CW2M	B575 N10276
Titanium Grade 2	B348 R50400	B367 C2	B348 R50400
Titanium Grade 3	B348 R50550	B367 C3	B348 R50550
Zirconium Grade 702	B550 R60702	B752 702C	B550 R60702
Zirconium Grade 705	B550 R60705	B752 705C	B550 R60705
11% Aluminum Bronze	---	B148 C95400	---

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Table 5 - Ratings for ASTM A126 Gray Cast Iron per ASME/ANSI B16.1-1989

Standard Class Working Pressure by Class, psig

Temperature, °F	Class 25 (see notes)		Class 125 (see notes)				Class 250 (see notes)				Class 800 (see notes)
	A126 Cl. A		A126 Cl. A	A126 Cl. B			A126 Cl. A	A126 Cl. B			A126 Cl. B
	NPS 4-36	NPS 42-96	NPS 1-12	NPS 1-12	NPS 14-24	NPS 30-48	NPS 1-12	NPS 1-12	NPS 14-24	NPS 30-48	NPS 2-12
-20 to 150	45	25	175	200	150	150	400	500	300	300	800
200	40	25	165	190	135	115	370	460	280	250	...
225	35	25	155	180	130	100	355	440	270	225	...
250	30	25	150	175	125	85	340	415	260	200	...
275	25	25	145	170	120	65	325	395	250	175	...
300	...	...	140	165	110	50	310	375	240	150	...
325	...	...	130	155	105	...	295	355	230	125	...
353	...	...	125	150	100	...	280	335	220	100	...
375	...	...	...	145	...	...	265	315	210	...	...
406	...	...	...	140	...	...	250	290	200	...	...
425	...	...	...	130	...	...	...	270	...	...	...
450	...	...	...	125	...	...	...	250	...	...	...

Notes:
Class 25. When Class 25 cast iron flanges and flanged fittings are used for gaseous service, the maximum pressure shall be limited to 25 psig. Tabulated pressure-temperature ratings for Class 25 cast iron flanges and flanged fittings are applicable for non-shock hydraulic service only.
Class 250. When used for liquid service, the tabulated pressure-temperature ratings in NPS 14 and larger are applicable to Class 250 flanges only and not to Class 250 fittings.
Class 800. The tabulated rating is not a steam rating and applies to non-shock hydraulic pressure only.
NPS is nominal pipe size.

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	Table 6 - Ratings for ASTM A395 Ductile Iron per ASME B16.42-1987	
	Standard Class Working Pressure by Class, psig	
Temperature, °F	Class 150	Class 300
-20 to 100	250	640
200	235	600
300	215	565
400	200	525
500	170	495
600	140	465
650	125	450

Notes:
Ratings are maximum allowable non-shock working pressures.

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Table 7
Ratings for Group 1.2 Materials
per ASME B16.34-1988 Table 2-1.2

	A203 B (a) A203 E (a)	A216 WCC (a) A350 LF3 (d)		A352 LC2 (d) A352 LC3 (d)		A352 LCC (e) A106 C (f)		
	Standard Class							
	150	300	400	600	900	1500	2500	4500
Temperature, °F	Working Pressure by Class, psig							
-20 to 100	290	750	1,000	1,500	2,250	3,750	6,250	11,250
200	260	750	1,000	1,500	2,250	3,750	6,250	11,250
300	230	730	970	1,455	2,185	3,640	6,070	10,925
400	200	705	940	1,410	2,115	3,530	5,880	10,585
500	170	665	885	1,330	1,995	3,325	5,540	9,965
600	140	605	805	1,210	1,815	3,025	5,040	9,070
650	125	590	785	1,175	1,765	2,940	4,905	8,825
700	110	570	755	1,135	1,705	2,840	4,730	8,515
750	95	505	670	1,010	1,510	2,520	4,200	7,560
800	80	410	550	825	1,235	2,060	3,430	6,170
850	65	270	355	535	805	1,340	2,230	4,010
900	50	170	230	345	515	860	1,430	2,570
950	35	105	140	205	310	515	860	1,545
1000	20	50	70	105	155	260	430	770

Notes:
(a) Permissible, but not recommended for prolonged usage above about 800°F.
(d) Not to be used over 650°F.
(e) Not to be used over 700°F.
(f) Not to be used over 800°F

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Table 8 - Ratings for Group 1.10 Materials per ASME B16.34-1988 Table 2-1.10								
	A182 F22 (c)		A217 WC9 (j)		A387 22 Cl. 2 (c)		A739 B22 (c)	
	Standard Class							
	150	300	400	600	900	1500	2500	4500
Temperature, °F	Working Pressure by Class, psig							
-20 to 100	290	750	1,000	1,500	2,250	3,750	6,250	11,250
200	260	715	955	1,430	2,150	3,580	5,965	10,740
300	230	675	905	1,355	2,030	3,385	5,640	10,150
400	200	650	865	1,295	1,945	3,240	5,400	9,720
500	170	640	855	1,280	1,920	3,200	5,330	9,595
600	140	605	805	1,210	1,815	3,025	5,040	9,070
650	125	590	785	1,175	1,765	2,940	4,905	8,825
700	110	570	755	1,135	1,705	2,840	4,730	8,515
750	95	530	710	1,065	1,595	2,660	4,430	7,970
800	80	510	675	1,015	1,525	2,540	4,230	7,610
850	65	485	650	975	1,460	2,435	4,060	7,305
900	50	450	600	900	1,350	2,245	3,745	6,740
950	35	380	505	755	1,130	1,885	3,145	5,660
1000	20	270	355	535	805	1,340	2,230	4,010
1050	20 (1)	200	265	400	595	995	1,660	2,985
1100	20 (1)	115	150	225	340	565	945	1,700
1150	20 (1)	105	140	205	310	515	860	1,545
1200	20 (1)	55	75	110	165	275	460	825

Notes:
(c) Permissible, but not recommended for prolonged usage above about 1100°F.
(j) Not to be used over 1100°F.
(1) For welding end valves only. Flanged end ratings terminate at 1100°F.

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	Table 9 - Ratings for Group 2.2 Materials per ASME B16.34-1988 Table 2-2.2							
	A182 F316 A182 F316H A240 316		A240 317 A240 316H A351 CF3A (d)		A351 CF3M (g) A351 CF8A (d) A351 CF8M		A479 316 A479 316H	
	Standard Class							
	150	300	400	600	900	1500	2500	4500
Temperature, °F	Working Pressure by Class, psig							
-20 to 100	275	720	960	1,440	2,160	3,600	6,000	10,800
200	240	620	825	1,240	1,860	3,095	5,160	9,290
300	215	560	745	1,120	1,680	2,795	4,660	8,390
400	195	515	685	1,030	1,540	2,570	4,280	7,705
500	170	480	635	955	1,435	2,390	3,980	7,165
600	140	450	600	905	1,355	2,255	3,760	6,770
650	125	445	590	890	1,330	2,220	3,700	6,660
700	110	430	575	865	1,295	2,160	3,600	6,480
750	95	425	565	845	1,270	2,110	3,520	6,335
800	80	415	555	830	1,245	2,075	3,460	6,230
850	65	405	540	810	1,215	2,030	3,380	6,085
900	50	395	525	790	1,180	1,970	3,280	5,905
950	35	385	515	775	1,160	1,930	3,220	5,795
1000	20	365	485	725	1,090	1,820	3,030	5,450
1050	20 (1)	360	480	720	1,080	1,800	3,000	5,400
1100	20 (1)	325	430	645	965	1,610	2,685	4,835
1150	20 (1)	275	365	550	825	1,370	2,285	4,115
1200	20 (1)	205	275	410	620	1,030	1,715	3,085
1250	20 (1)	180	245	365	545	910	1,515	2,725
1300	20 (1)	140	185	275	410	685	1,145	2,060
1350	20 (1)	105	140	205	310	515	860	1,545
1400	20 (1)	75	100	150	225	380	630	1,130
1450	20 (1)	60	80	115	175	290	485	875
1500	15 (1)	40	55	85	125	205	345	620

Notes:
(d) Not to be used over 650°F.
(g) Not to be used over 850°F.
(1) For welding end valves only. Flanged end ratings terminate at 1100°F.

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